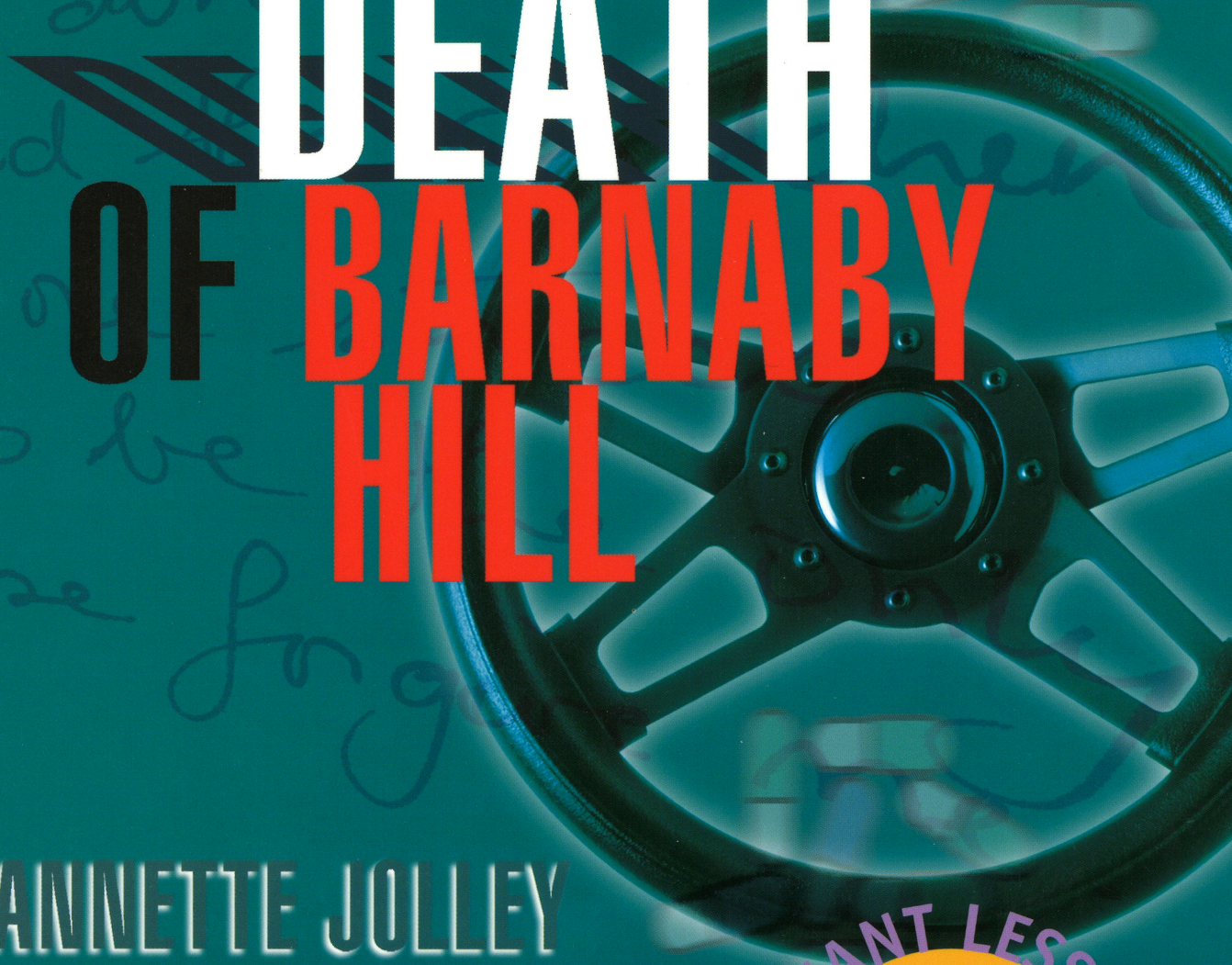
FORENSIC SCIENCE HYPOTHETICAL

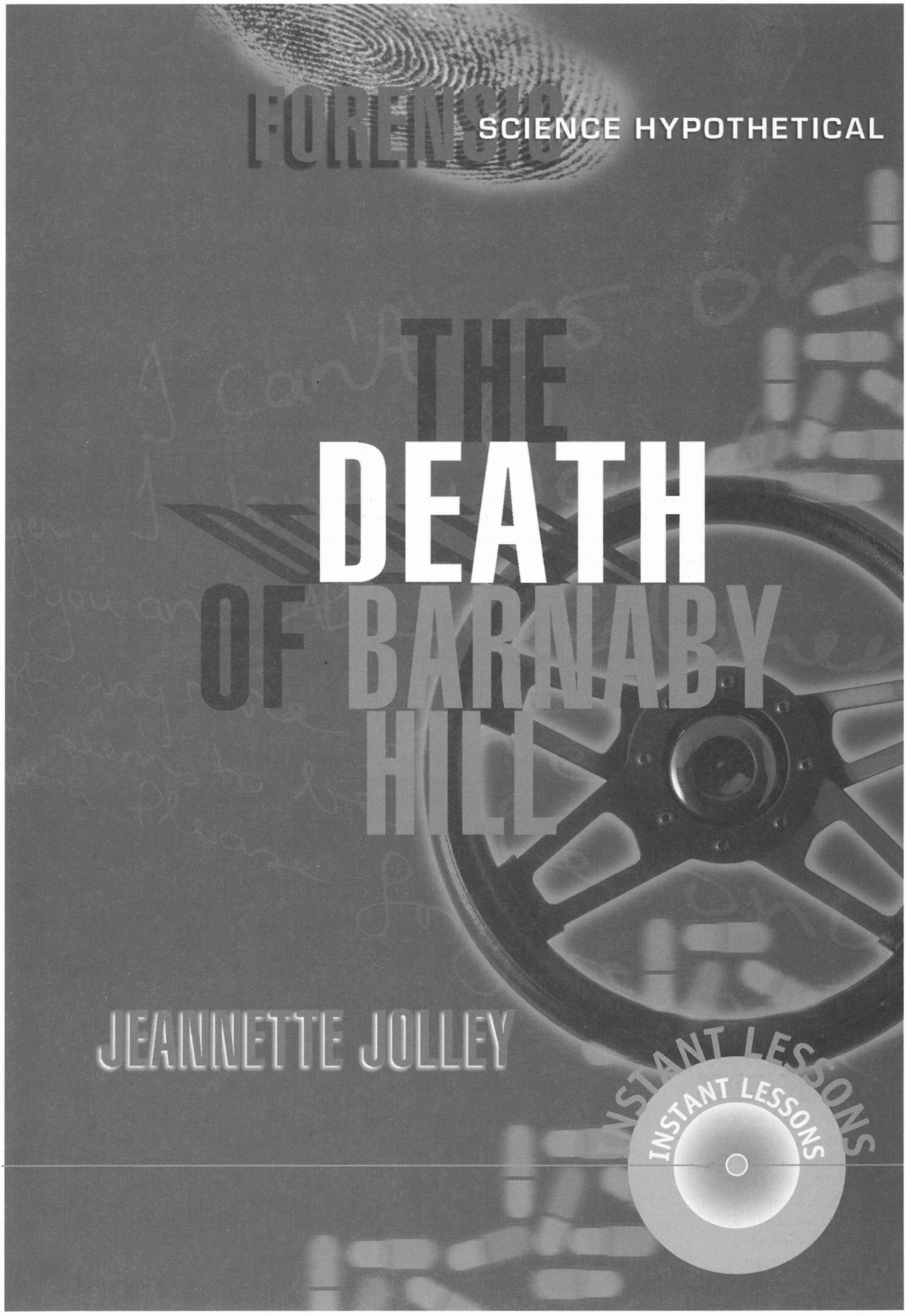
THE DEATH OF BARNABY HILL



JEANNETTE JOLLEY

INSTANT LESSONS

Quality ready-to-use resources


FORENSIC SCIENCE HYPOTHETICAL
THE DEATH OF BARNABY HILL
JEANNETTE JOLLEY
INSTANT LESSONS

Jeannette Jolley (B.Sc., Dip.Ed., B.Ed.) is a science teacher at St Clare's College in the Australian Capital Territory. She has over 29 years of experience teaching science in three different states. This includes ten years of teaching forensic science as a semester unit. Jeannette was awarded the inaugural ACT Westfield Chief Minister's Scholarship in 2002 for a science teacher to travel overseas to undertake research or a course of study. She used it to complete a Volume Crime Scene Examiner's Course with the UK police at their National Training Centre in County Durham.

Other titles by Jeannette Jolley

- Forensic Science Hypothetical
  - The Kidnapping of Jamie Lawson
  - The Burglary in Sutton Street
- Forensic Science for High Schools (Books 1–4)
- Everyday Science (Books 1–2)

© Jeannette Jolley and Blake Education, 2006

Downloadables sold with a licence to print and photocopy
The contents of this book are covered by copyright laws in Australia and in other countries.
The purchase price includes the right for you to print and photocopy these pages for your students without further payment to the publisher. For further information about how schools can use this photocopiable resource, please refer to the 'FAQ's on www.blake.com.au.
Otherwise, you may only download, print, copy, store and communicate the contents of this book without permission when that is allowed by the copyright laws applicable to you.
If you work in an Australian school that is covered by the education statutory licence, you may be allowed to copy, download, print, store and communicate parts of this book without permission from copyright owners. Under the education statutory licence, payments are made on behalf of schools to Copyright Agency, which distributes the payments to writers, artists and publishers. See www.copyright.com.au.

Copyright Agency
Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: +612 9394 7600
Facsimile: +612 9394 7601
E-mail: memberservices@copyright.com.au

Commissioning editor: Jennifer Gough
Managing editor: Ruth Schultz
Editor: Ruth Schultz
Page formatting: Peter Marshall
Proofreader: Judith Schultz
Illustrations: Boris Silvestri
Cover design: Gail McManus Graphics
Printed in Australia by Ligare Book Printer

Blake EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au

ISBN 978 1 92561 088 8

Introduction

The Death of Barnaby Hill is a practical extended project in which students are the detectives who must solve the crime. They will enjoy this 'real-life' application of skills and knowledge introduced in Jeannette Jolley's series *Forensic Science for High Schools Books* (Books 1–4). However, solving this crime does not require any previous study or formal knowledge of forensic science.

This book, like *The Kidnapping of Jamie Lawson*, is aimed at middle secondary level, but is also suitable for Year 8 students who are looking for a challenge. It can also be part of a non-tertiary senior science unit. These hypothetical crimes have been used as practical assessment tasks after forensic science has been taught formally as a topic, as well as an engaging exercise to fill in a gap of several weeks when core work has been completed.

Teacher preparation

Exercises 1, 2 and 3 are offered as three single introductory lessons for students who have no formal knowledge or understanding of forensic science. They can be done before undertaking the solving of the hypothetical crime in Parts 1 to 7.

Exercise 1: Recording shoe prints and tyre marks After students complete the exercise, you could have the class make their own shoe prints in sand, and then make casts using plaster of paris.

Exercise 2: Inspecting hairs and fibres After the exercise, students could examine under a microscope their own hair, and then hair taken from other people and animals.

Exercise 3: Using chromatography to compare inks Students can practise separating out the pigments by using different coloured water-soluble, felt-tipped pens or overhead markers. Food dye also works very well. See page 34 for detailed instructions.

Exercises on searching a crime scene, fingerprinting and handwriting comparisons are offered in a previous book in this series, *The Kidnapping of Jamie Lawson*.

How to prepare the evidence

The following 'evidence' is needed for Part 5. The quantities of hair and fibres are for one set only. However, for classes of over 20 students, at least two sets are recommended.

- Five small press-seal bags with stick-on labels (10 for two sets)
- Four types of wool or synthetic yarn, all in a dark blue colour. A strand of each sample as described in the table below, should be placed in each of the 5 press-seal bags.

Evidence	Label
Fibres from first 'wool' sample	Taken from jumper belonging to Brian McDonald
Fibres from second 'wool' sample	Taken from jumper belonging to Timothy Hill
Fibres from third 'wool' sample	Taken from jumper belonging to Helena Hill
Fibres from fourth 'wool' sample	Taken from jumper belonging to Steve Nara
Fibres from first 'wool' samples	Fibres found attached to cloth at crime scene

- One sheet of A4 paper
- Four strands of different shades of brownish hair, about 25 cm long, taken from four different people, plus a second strand of hair from the first person. These five strands should be taped lengthwise, side by side, to the A4 paper, and labelled as shown in the table on below.

Hair from first person	Hair taken from Sarah Hill
Hair from second person	Hair taken from Helena Hill
Hair from third person	Hair taken from Isabella McDonald
Hair from fourth person	Hair taken from Amy Wallace
Hair from first person	Hair found on seat belt of car at crime scene

- Five different brands of black felt-tipped pens, each labelled with one of the following names: Helena Hill, Isabella McDonald, Sarah Hill, Timothy Hill, Steve Nara (only one set per class required)

© Blake Education 2006

ISBN 978 1 92561 088 8

- Several magnifying lenses or microscopes (aim for one per group)
- Six strips of chromatography paper, a medium beaker and water (one set per group)
- One copy of each of the two notes per class. The suicide note should be copied by the teacher using the pen labelled 'Helena Hill' and the second note copied using the pen labelled 'Isabella McDonald'. Each group should be given one word cut from each note to test. For authenticity in handwriting style, you could trace the notes supplied in the evidence.
- 3 m of garden hose plus two pieces of a coloured cloth (for realism, but optional)
- common kitchen knife (optional, tell students fingerprints have already been recorded).

Summary of the hypothetical crime

Part 1 – A body is discovered (pages 8–9)

Is it murder or suicide? A young man discovers an elderly neighbour dead in his car with the engine running. There is a piece of garden hose attached to the exhaust pipe and inserted into the car through a back window. The police arrive and find a suicide note.

Part 2 – The scene is searched at first light (pages 9–11)

Initial physical evidence found in the garden is introduced with maps and diagrams that show what was found where. Students take on the role of detective and are asked to explain what they would like to do with the evidence.

Part 3 – Information obtained by late morning (pages 12–15)

This includes statements from the victim's wife and neighbour who found the body, as well as evidence found when a more thorough search was undertaken inside the Hill house. Students are asked to summarise what they have found out so far and record what other information they need to seek out next.

Part 4 – Information obtained by late afternoon (pages 16–33)

This includes the post mortem report, handwriting samples and fingerprints from suspects. Students compare the fingerprints found at the crime scene and handwriting on the two notes with samples taken from suspects. More witness statements and additional information give the students further leads to follow.

Part 5 – Some practical activities (page 34)

Students use chromatography to test inks from the suspects' pens and the notes, to check if they are similar. Magnifying lenses are used to compare hair and fibre samples to discover who might have left them at the crime scene.

Part 6 – Information available early next day (pages 35–38)

Students compare shoe and tyre prints from the suspects with those found at the crime scene. These, with the statements, start to bring the whole scenario together.

Part 7 – Final evidence from a suspect and the lab (pages 39–40)

There is now enough information to determine how the crime was carried out, by whom and why.

Teaching instructions

Each part of the hypothetical crime and the sets of elimination fingerprints should be photocopied separately on different coloured paper, and stapled together. In this way, the different parts are easily recognised and can quickly be returned to the correct pile at the end of a lesson. All of the sheets can be reused with several classes. Students are able to work through each part at their own rate. However, earlier parts should be available so that information can be rechecked. All seven parts can be worked on in groups. Two to three students appears to be the best group size to ensure each student stays involved and has a role.

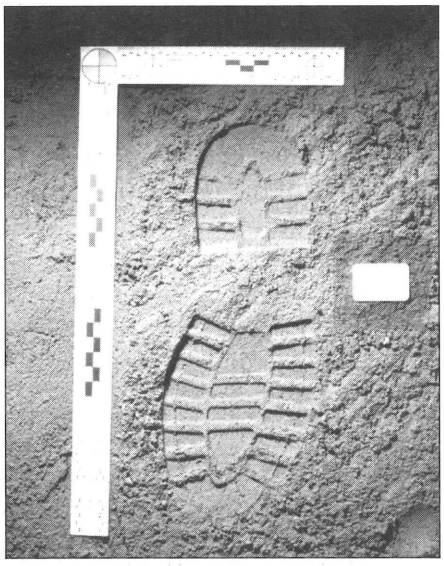
A photocopiable report for each group to record its progress is provided on pages 41–46. If the report pages are copied single sided, students will have enough space for any additional information they may wish to note. You may prefer to have a different scribe complete each part of the report. Some classes develop a competition between the groups, aiming to be the first to complete the report and solve the crime correctly.

If using this exercise as an assessment task, Part 1 may be done individually for homework or in class. Students could be asked to discuss what the neighbour should do after he found the body, and what the police should do to gather evidence and discover the possible suspects. The students could be asked to suggest the most important questions the police should ask Mrs Hill (Barnaby Hill's widow) and her neighbour Patrick.

© Blake Education 2006
ISBN 978 1 92561 088 8

Recording shoe prints and tyre marks — ACTIVITY 1

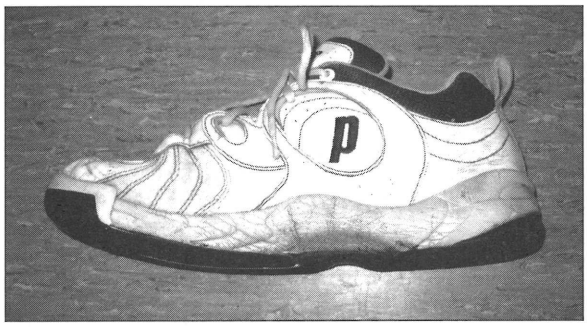
Name _____



It is amazing how often impressions are left at a crime scene. They might be from the tyres of a 'get-away' vehicle, or from a criminal's shoes. Even the tools used to break into a building can leave impressions on a windowsill. When pressed on by an object, soft garden soil, mud or timber will remain in the shape it was forced into by that object – and so leave an impression.

When an impression is found, it is photographed with a strong light source coming from one side. This ensures that the ridges and depressions show up clearly. The photo on the right was taken with light coming from the left, and you can see the small shadow that is cast near the edges of the impression. A ruler is included in the photo so that the exact size of the shoe is clear, even if the photo is later enlarged.

As well as taking a photograph, crime scene examiners make a cast of the impression, using a substance like plaster of Paris if the impression is large. The plaster mixes quickly and easily to a smooth paste that is poured into the impression. For smaller impressions such as tool marks, silicone rubber is used instead of plaster. After a short time, the investigators working on the crime will have a three-dimensional copy of the object that made the impression. If they then find a similar object, such as a shoe belonging to a suspect, they can compare it to the photograph or cast to see if they match.



1 Does the shoe in the photographs above match the shoe print at the top of the page? Explain your answer.

2 Why are photographs and casts made of impressions found at a crime scene?

3 Why do you think crime scene examiners both take photographs and make casts?

4 Do you think that matching a shoe with an impression found at a crime scene is enough information to convict the shoe's owner of a crime committed at that site? Give reasons for your answer.

5 What other information – besides a shoe print – would be useful when trying to work out if a suspect is guilty of committing a crime? Explain why.

© Blake Education 2006

ISBN 978 1 92561 088 8

Inspecting hairs and fibres ACTIVITY 2

Name _____

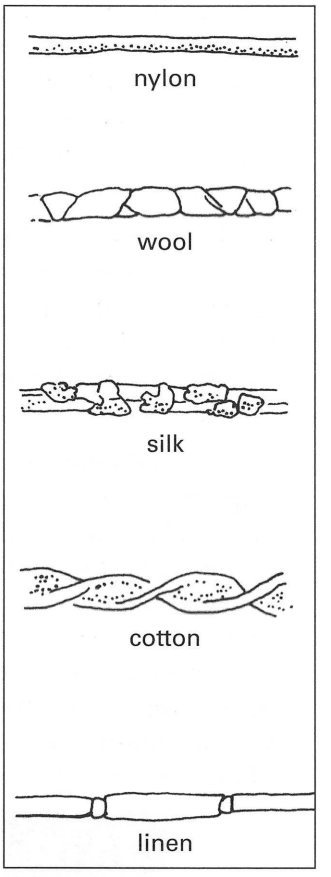
Some of the most common examples of trace evidence left behind at a crime scene include hairs from a criminal's body and fibres shed from their clothes. This is why hairs and fibres are searched for, very carefully. As they are so small, a criminal would not even notice that they had been left behind.

A forensic scientist who specialises in hair identification can discover from which type of animal the hair came, and – if the hair is human – which part of the body the hair came from. Microscopic examination can indicate whether the hair was pulled out, cut off or just fell out naturally. Chemical tests can sometimes reveal any drugs or poisons the person took, any hairdressing treatment like dyes and bleaches or in some cases, even the foods eaten by the person. If there are any cells on the hair roots, then a DNA profile can be made from the cells and the person identified if their profile is on police records.

Fibres are classified as either synthetic (man-made), such as nylon and polyester, or natural. If natural, they might come from plants such as cotton or from animals such as sheep (wool) or caterpillars (silk).

Synthetic fibres are usually produced from a syrup-like liquid that dries into a long thin filament of uniform thickness and colour. These filaments can be spun together into thread. In the laboratory, samples can be compared using chemical tests to see if they are made of exactly the same substances. Strength comparisons can be made by measuring the force needed to break the thread, or melting points can be compared.

Natural fibres are far less uniform in their appearance. Their quality depends on the health of the animal or plant and even on the seasons in which it grew. To identify the type of natural fibre, inspection under a microscope is usually done first. Dyed samples can also be matched using ultra-violet (UV) light, which shows how both samples reflect this light and if this reflection is exactly the same.



1 List all the information an expert in hair analysis could find out from a single hair found at a crime scene.

2 In what way might rabbit hair found at a crime scene be useful evidence in helping to solve a crime?

3 a What are the two basic groups that all fibres can be divided into?

b Use a dictionary or the Internet to identify which of the groups you listed above these fibres belong to: polyamide, angora, vinyl, alpaca, acrylic, linen and mohair. Where appropriate, say whether the fibre comes from a plant and what type of plant, or from an animal and what type of animal.

4 List the tests that fibres undergo to determine whether two similar fibres came from the same source.

5 If a fibre found at a crime scene was identical to fibres from a suspect's clothing, does that prove the suspect was at the crime scene? What other information is needed to prove the guilt of the suspect? Explain your answer.

© Blake Education 2006
ISBN 978 1 92561 088 8

Using chromatography to compare inks ACTIVITY 3

Name _____

The inks found in pens are made from a number of different chemicals. These chemicals are combined to give the desired colour and consistency needed to leave a clear line on a page of paper. Different manufacturers use slightly different combinations of these chemicals to produce their inks, so each brand of pen has its own unique formula.

The chemicals in different inks can be compared using the simple technique of chromatography. A sample of the ink is placed about 1 cm from the bottom of a piece of absorbent paper, which is then stood in a beaker that contains a small amount of solvent, such as water or methylated spirits. The solvent slowly soaks up the paper. As it meets the ink, some of the chemicals in the ink will dissolve and be carried up with the solvent. This leaves a trail of different colours spread up the paper.

Some of the chemicals consist of very small molecules, and these will be carried furthest up the paper, especially if they are very soluble. Where the ink chemicals are made of larger molecules that are less soluble, these will not be carried as fast or as far up the paper. Each ink usually leaves its own unique pattern of colours, called a **chromatogram**. If two chromatograms have the same colours in the same order, it is highly likely that both samples came from a pen of the same type and brand.

Today in the forensic laboratory, chromatography is not used as described above. Instead, scientists use instruments that measure the exact chemicals found in a mixture. A small sample of the substance is placed in a chamber where it is heated so that it evaporates as a gas. The types of particles in the gas are identified, and the scientist can print out a graph that shows the type and amount of each particle in the sample.

1 Describe how a chromatogram is made.

2 Explain why some colours move further up the paper than other colours.

3 If two chromatograms are found to be identical, does that mean that they were made with exactly the same pen? Give reasons for your answer.

4 When a student was comparing two chromatograms, she found that they showed exactly the same colours in the same order, but one had its colours spread further up the paper than the other. Is it possible that they were written using the same pen? Explain your answer.

5 If a chromatogram that was produced using a suspect's pen matched a chromatogram produced from a threatening note, does it mean that the suspect wrote the note and should be charged with the crime? Explain your answer.

6 Would you expect forensic investigators today to make a chromatogram in exactly the same way as described above if they were comparing two inks? Explain your answer.

7 Try the above technique using different coloured paint or water-soluble pens. Draw a large dot about 1 cm from the bottom of a strip of blotting or filter paper, one strip for each pen or paint. Stand the strips upright in a beaker or glass containing about 0.5 cm of water. Watch the colours move up the paper as the water seeps upwards.

© Blake Education 2006

ISBN 978 1 92561 088 8

A body is discovered .. .. PART 1

Use the information in this part to fill in Part 1 of your report.

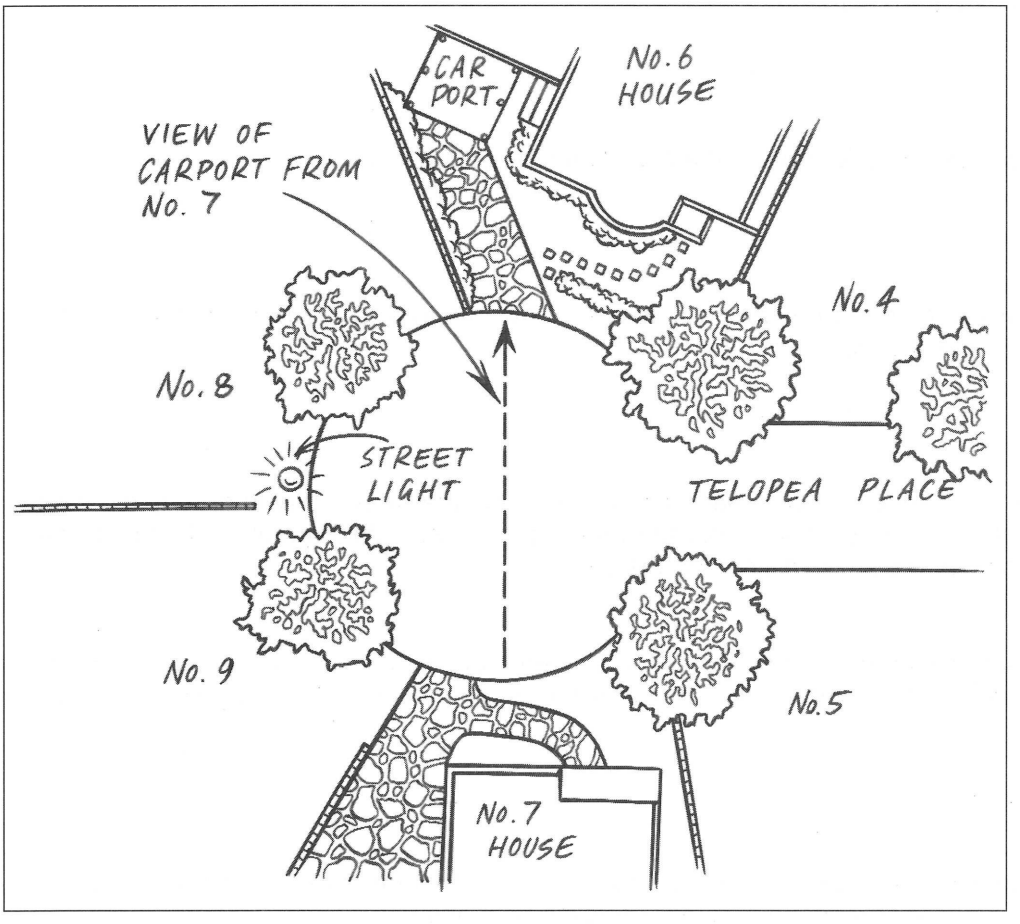
Patrick Sullivan woke up and realised that he could still hear the humming sound of a car engine from his dream, even though he was now awake. He glanced over at the numbers glowing in the dark from his clock radio. It said 3.17, which meant he still had about four hours before he had to get up. He rolled over to go back to sleep, but the car engine continued its annoying drone. Patrick wondered who it was. Most of his neighbours were older retired couples who tended to go to bed early. They were not in the habit of hosting visitors who kept their car engines running in the early hours of the morning.

Patrick crawled out of bed to take a look. His bedroom was at the front of the house and gave a good view of the small street. He could see the Volvo belonging to Barnaby Hill, his sixty-year-old neighbour, parked in the driveway opposite at number 6. There were no other cars in sight and the noise seemed to be coming from that direction. Patrick thought he could see a person sitting in the front seat. It seemed quite odd.

Annoyed that his sleep had been disturbed, Patrick reached over and pulled on his tracksuit. He put on some shoes and left his house to investigate.

The Hill house, like every other house in the small cul-de-sac, was in darkness. Patrick wondered where Mrs Hill was as he strode across the street to see why Barnaby was disturbing the peace at this time of the night. The automatic security lights came on as he stepped onto the Hill driveway. He pulled open the car door, and was hit with a cloud of exhaust fumes as the body of Barnaby Hill slumped towards him. Patrick's reflex reaction caught the body as it fell. He carefully laid Barnaby's body on the path beside the car. As he reached in and switched off the car's ignition, he noticed a garden hose coming from the exhaust pipe. It was pouring poisonous gases from the car's exhaust through a small gap in a rear window of the car, behind where Barnaby had been sitting. It looked like Barnaby Hill had decided to commit suicide.

Map of houses in Telopea Place, Benwarra



© Blake Education 2006
ISBN 978 1 92561 088 8

Information found during early hours of morning

Mrs Hill was difficult to wake when Margaret Collins, a police officer, first entered the bedroom to inform her of Barnaby's death. Once awake, she became quite hysterical. Dr Samuel Simpson, the pathologist (medical examiner) tending her husband, was summoned upstairs to calm her down. Her son Timothy and daughter Sarah were both contacted.

While waiting for Sarah and Timothy to arrive, Mrs Hill told Officer Collins that she went to bed about 10 pm, after taking some sleeping tablets. She had left Barnaby downstairs talking to their son Timothy and his girlfriend Amy at about quarter to ten.

Dr Simpson made an initial examination of Barnaby's body before sending it off for a detailed post mortem (autopsy) later in the day. All he was able to tell the police at this stage was that Barnaby Hill was deceased and that his appearance was consistent with carbon monoxide (exhaust gas) poisoning.

Paul Anderson was one of the first police officers to arrive at the scene. While he watched the doctor examine the body, he took note of the surroundings and recorded his observations. He noted that Barnaby Hill was wearing pyjamas and had bare feet, which were quite clean. One slipper, which Mrs Hill later identified as belonging to her husband, was found on the path between the side door and the carport. Its mate was found in the kitchen near the side door.

The house had been in darkness but not locked when the police arrived. There were no signs of any disturbance and no one had heard anything strange or unusual until the sound of the car engine woke Patrick Sullivan. A suicide note was found, folded in half and addressed to Helena (Mrs Hill). It had been placed propped up on a shelf in a glass cabinet in the kitchen (see copy at right).

Sorry Helena. I can't go on any longer. I don't want to put you and the children through any more stress. This seems to be the only option. Please forgive me.

The scene is searched at first light PART 2

Use the information in this part to fill in Part 2 of your report.

Mrs Hill was too distraught and confused to be formally questioned when she was first woken by the police. However, once her daughter Sarah arrived, she agreed to have a shower and get dressed so that she could help the police find out what had happened to her husband. Timothy, Sarah's brother, suggested his mother be brought to his house where she could have breakfast and then be interviewed when she felt up to it. The police cordoned off the Hill house, garden and driveway with crime scene tape. Once Mrs Hill had left and it was light, the forensic team made a more thorough search of the house and gardens.

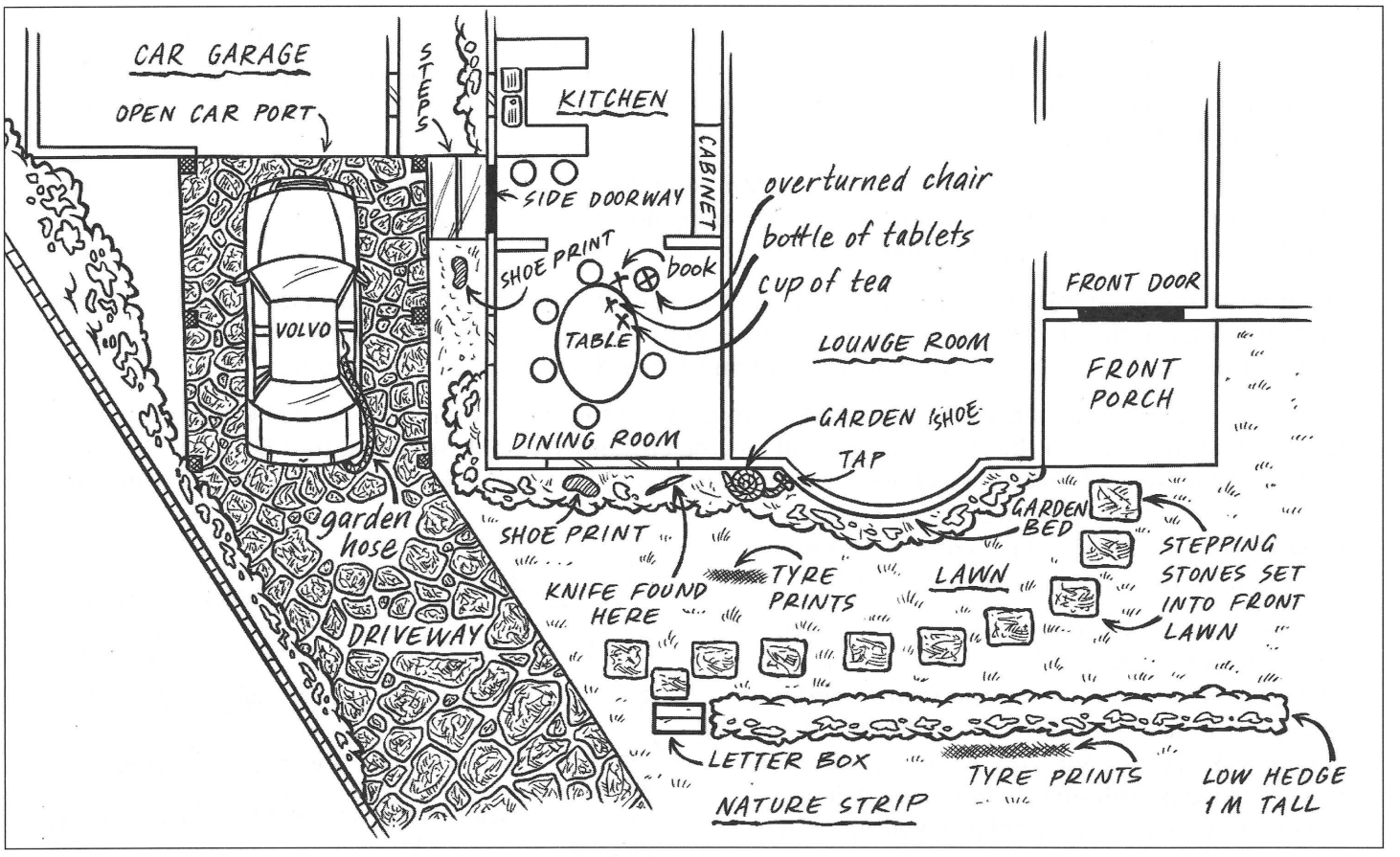
Evidence found by crime scene examiners after dawn

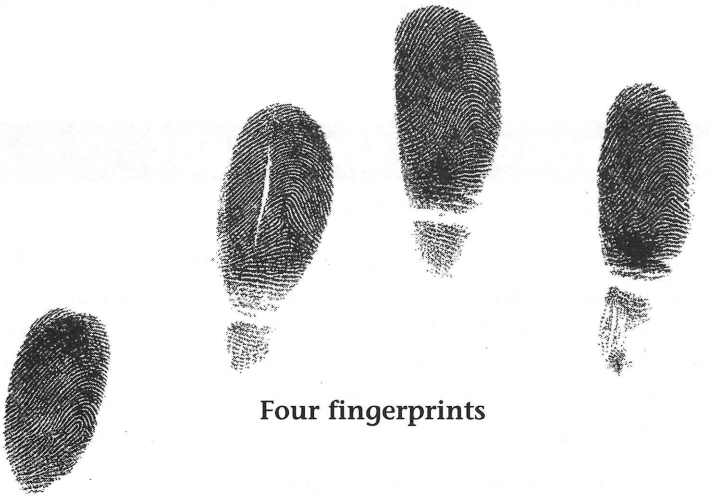
- 3 m of garden hose pushed into a rear window of the car, held in place and sealed with some old cloth. More of this cloth tied the other end to the exhaust pipe.
- Some dark blue fibres, caught in the cloth that had been tied to the exhaust pipe
- An old garden hose found near the front garden tap of the Hill house
- A knife found in the garden near the tap. Fingerprints were found on the handle of this knife.
- Four fingerprints and a dark brown stain, which tested positive to a presumptive blood test, found near the driver's door handle (see diagram on page 11)
- Three fingerprints and a thumb print on the back of the car (see diagram on page 11)
- Two brown hairs approximately 25 cm long, caught in the driver's seat belt
- A shoe print found in the garden bed outside the dining room window. A different shoe print found in a garden bed at the front of the house
- Tyre prints found on the front lawn and others with a different pattern found on the nature strip
- Hand print on glass doors of the kitchen cabinet, where the suicide note was found (see diagram on page 10)

© Blake Education 2006

ISBN 978 1 92561 088 8

Map of house and garden of 6 Telopea Place showing the location of evidence

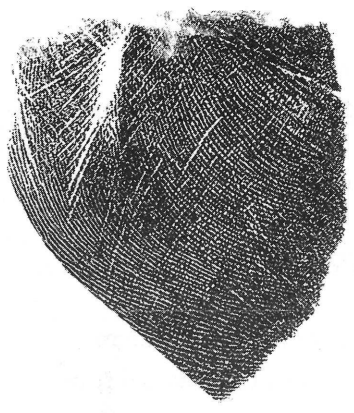


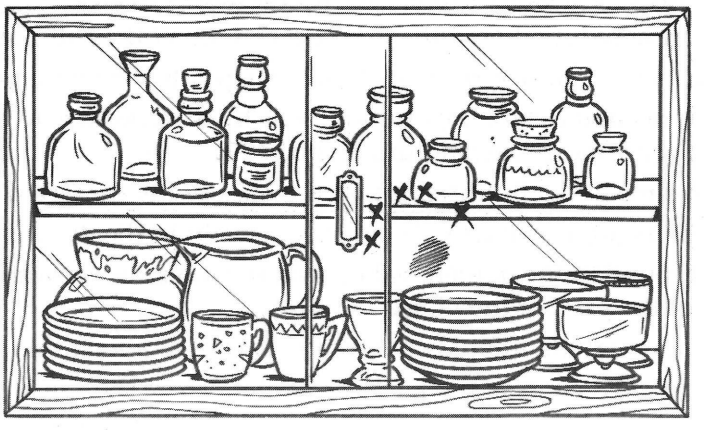


Four fingerprints

Thumb print

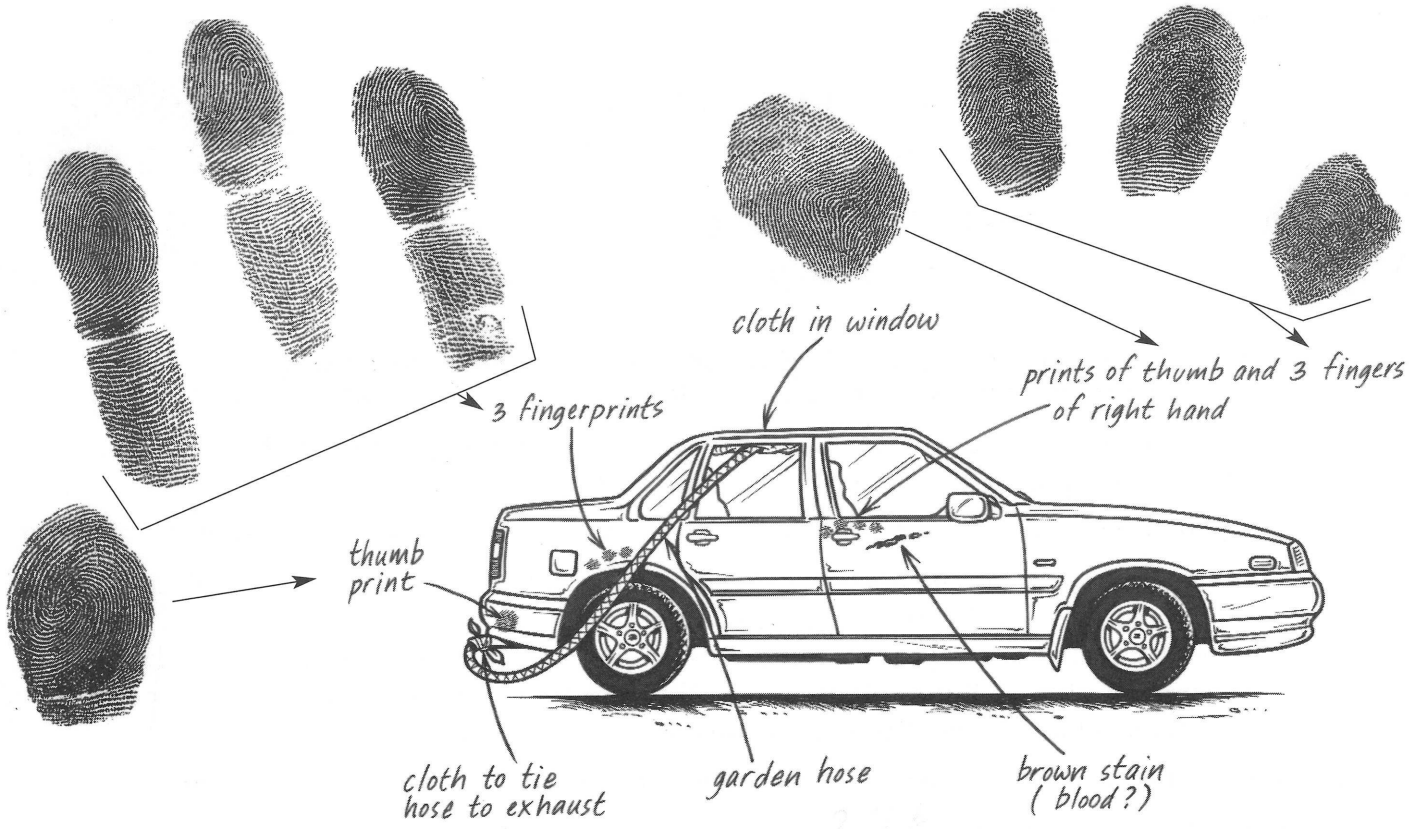
Partial palm print



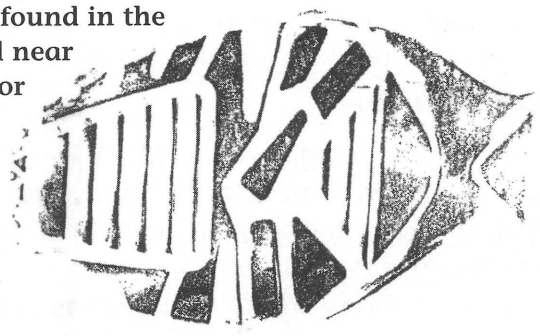


Artist's drawing of kitchen cabinet, with location of prints marked

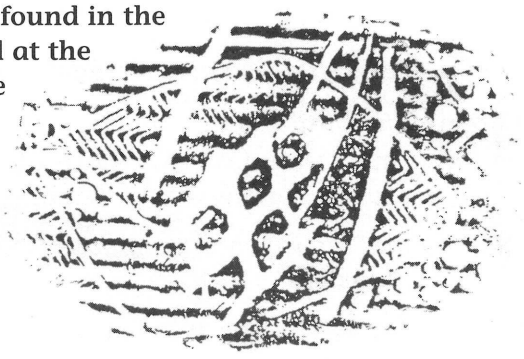
© Blake Education 2006
ISBN 978 1 92561 088 8



Shoe print found in the garden bed near the side door

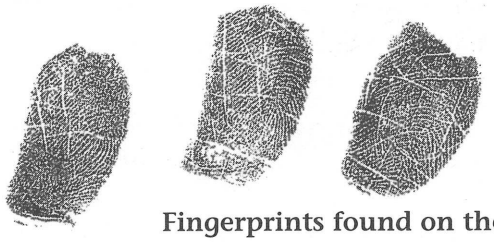


Shoe print found in the garden bed at the front of the house

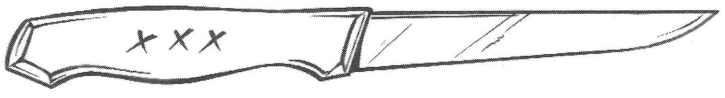


Tyre prints found on the front lawn

Tyre prints found on the nature strip



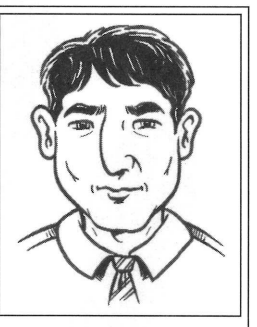
Fingerprints found on the knife



ISBN 978 1 92561 088 8

Information obtained by late morning PART 3

Use the information in this part to fill in Part 3 of your report.



POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Patrick Sullivan

Taken by P.C. Collins Date 22.05.10 Time 8.15 am

Q Thank you for coming in to help us Mr Sullivan. For the benefit of the tape, please give your full name, age, address, occupation and your relationship to the deceased.

My name is Patrick Wayne Sullivan. I am 33 years of age, and an accountant. I live at 7 Telopea Place, Benwarra, opposite the Hill house. So Barnaby Hill is my neighbour.

Q Thank you. Now please, in your own words, explain what happened when you woke this morning.

Well, I must've been dreaming about a car engine. Then, when I woke up and I could still hear it running, I realised it was for real. At first I thought one of our neighbours had a late visitor as it was after 3 am, but it just kept going. After listening for five minutes it was really annoying, so I got up to see who it was. Telopea Place is a really quiet street normally. Most of the residents are older, so you don't get many disturbances like that.

Q Yes, I see. Can you describe what you saw when you got up?

Well I looked out the window and it was quite dark, except for one streetlight at the end of the cul de sac and one at the top where it meets Banksia Avenue. The Hill house is opposite mine, and I could see Barnaby Hill's car in the carport. The sound seemed to be coming from there. It was too dark to see who was in the car, so I decided to throw on some clothes and check it out.

Q Tell me exactly what you saw as you approached the Hill house.

As soon as I stepped outside it was clear that the sound was coming from Mr Hill's Volvo. His house was dark and there was no sign of life. As I walked up the drive, the security lights came on and I could see Barnaby sitting in the car. I wondered what he was doing so late at night. I opened the driver's door and was startled when Barnaby's body slumped towards me. I was lucky to catch him so he didn't fall right out. The smell of exhaust fumes was very strong. I turned the ignition off and pulled his body out onto the driveway. It felt cold and it was obvious from his bluish lips and staring eyes that he was dead, so I rang the police.

Q Thank you. Now tell me as accurately as possible when it was that you discovered Mr Hill's body.

Well it was 17 past 3 when I first woke up. It was about five minutes later, maybe more, when I got up to look out the window. Then it must have been five more minutes before I was dressed and went across the road to investigate. So that makes it about 3.30 I suppose.

Q Good. Your phone call was logged at exactly 3.34 am, so that fits in. I don't suppose you would know of any reasons why Mr Hill might take his own life, do you?

Not really. We aren't close. All I can think of is that he may've been worried about a recent rumour. It was suggested by some other neighbours that Barnaby Hill was being investigated for molesting a student about 20 years ago. They warned people to keep their children away from the Hills' house. I read something in the papers about it, I think, but there were no names mentioned. But the description of a local, recently retired school principal describes Barnaby Hill pretty well.

Q Well thank you for your help Mr Sullivan. We appreciate your coming in and helping us with our enquiries. If we have any further questions or details we need to recheck we'll be in touch.

This statement of 1 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed P Sullivan

© Blake Education 2006
ISBN 978 1 92561 088 8

POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Helena Hill

Taken by P.C. Collins Date 22.05.10 Time 9.00 am

Q Thank you very much Mrs Hill for agreeing to answer our questions so early in the morning at this difficult time. To help us find out what happened to your husband, we'd like to record our conversation, if that's OK with you? Firstly, for the benefit of the tape, could you please state your name, age, occupation, current address and your relationship to the deceased.

Yes. My name is Helena Katarina Hill and I am 57 years of age. I am a housewife and live at 6 Telopea Place. I am, or was, Barnaby Hill's wife. *[Her voice becomes somewhat broken, but she insists on continuing.]* Oh, it's so hard to get used to the idea that he's gone. I can't believe it, it's all so sudden. *[Mrs Hill was comforted by her daughter Sarah who was allowed to stay with her mother during the interview.]*

Q Mrs Hill you are being very brave, and we will try to get this over with as quickly as possible. This morning you mentioned having a dinner party at your house yesterday evening. Could you start from there and tell us exactly what you and your husband did yesterday?

Well it wasn't exactly a dinner party. Last week we invited our good friends Dr Brian McDonald and his wife Isabella over for a meal. Then on Friday our son Tim rang and said he and his girlfriend Amy wanted to come over and talk to Dad about something – they want to get engaged but Barnaby wasn't happy about it – so I invited them to come for the meal as well. I hadn't seen them for about a week. Then when Sarah arrived during the afternoon I asked her to stay as well.

Q So, Mrs Hill, you had five guests for dinner. Can you tell us when everyone arrived?

Yes, I think so. I'd asked everyone to come about 7. Barnaby must have told Brian 6.30, because they arrived early and I wasn't ready. So Barnaby took them into the den for some drinks while Sarah helped me finish the meal. She's so wonderful. Not like Isabella, who didn't offer to help, even when she knew she'd come too early. Isabella likes to be waited on, and prefers to join the men. She's a terrible flirt – you should see her trying to catch Barnaby's eye.

Q Mrs Hill, can you remember when your son and his girlfriend arrived?

Yes, it was right on 7. I've always brought up the children to be punctual. I looked at the clock when the doorbell rang and it was exactly 7 o'clock.

Q So you had a pleasant meal and there were no problems, interruptions or arguments?

No, not an argument exactly, but there was one interruption. During the meal we had a visit from our fencing contractor. It was rather upsetting and created some tension. We recently had a new fence put up by Steve Nara from Erindale. Barnaby refused to pay him because he said that the fence is not in the right place. Barnaby said that Steve built the fence 20 cm into our land. The Goughs next door have paid him for their share of the fence, but Barnaby won't pay unless the new fence is pulled down and put in the right place. Mr Nara argues that he put it where the old fence was, and that it was our responsibility to have the property boundary surveyed. Barnaby and I have had many disagreements about it. I told Barnaby to pay the man because he's paid for the materials and it's only a little out anyway. Sarah agrees with me, but Brian and Isabella both agree with Barnaby, and say it is the principle of the thing. Steve Nara told Barnaby that if he doesn't pay he'll take him to court. I suppose I'll have to pay him now that Barnaby's gone.

Q So that upset Barnaby last night?

Yes, but obviously not enough to kill himself! And he wasn't upset – he was angry.

© Blake Education 2006

ISBN 978 1 92561 088 8

Q *Of course. However, we have still to establish that it was suicide. Can you tell me what time Steve Nara visited, and what happened next?*

Well I suppose it was just after 8 o'clock when Steve knocked on the door, because we'd just finished our main course and I was clearing away the dishes. I went into the kitchen to get the desserts when I heard the doorbell ring. Barnaby didn't talk to him for long. I think he closed the door in the man's face. I heard Barnaby shouting and challenging the man to take him to court, when I was carrying the dessert tray into the dining room. After we finished dessert, Brian came out to help me get the coffee. That's most unusual, but I think he wanted to hurry me up. He went out to the car to get Isabella's sugar pills, then he helped carry out the coffee cups. He and Isabella left soon after they finished their coffee. In fact, I think Brian was upset with Isabella over something. You know, you can always sense tension, even if no words are said. Did you notice that too Sarah? *[Sarah agreed with her mother, that Dr McDonald seemed angry with his wife about something, but she had no idea what it was about.]*

Q *So about what time did they leave, would you say?*

It was about 9.30, because Brian said something about taking his wife home, then calling into the hospital to check up on a patient he was worried about. This was the first time he mentioned going home early.

Q *So then, can you tell us when the others left?*

Well let's see. Sarah helped me clear the table and stack the dishes, while Tim and Amy went into the den to speak to Barnaby. Sarah left about 15 minutes after the McDonalds left I would say. *[Sarah nodded in agreement with her mother regarding her estimates of time.]*

Q *So that left Timothy and Amy. Can you tell me when they left?*

No, I'm afraid not. I had a headache after the hassles over paying Steve Nara. I took some painkillers, then some sleeping tablets. With the worry over being taken to court and then some other worries, I've had some difficulty sleeping lately. So I went to bed and didn't actually hear them leave. In fact, the next thing I remember is being woken by the police.

Q *Mrs Hill, I have to ask you one more thing. You mentioned some other worries. Would that include the recent accusations of molestation by one of Barnaby's former students?*

Yes, I suppose you would know all about that. Barnaby has been very brave, standing up to that young man's accusations and insisting that what he's being accused of is a lie. He says that it's all been made up. But mud sticks he says, and he was worried that it would get into the papers and he'd lose his reputation among his colleagues and all our friends.

Q *So, do you think Barnaby might have been worried enough about that to commit suicide?*

No, absolutely not. Barnaby would see that as admitting guilt. He was going to fight it to the end he said. The young man, Lars Vogel, threatened to destroy Barnaby's life – how true that is now. Maybe he killed Barnaby, you did say that it might not be suicide. Do you think Lars could be the murderer?

Q *Mrs Hill, we are following all leads to get to the answer as quickly as possible. We have a lot of experienced people working on the case. You have been most helpful, and Sarah too. It's been most appreciated. We'll keep in touch and let you know of our progress. We may have some laboratory results to share with you by this afternoon.*

This statement of 2 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed Helen Hill

© Blake Education 2006

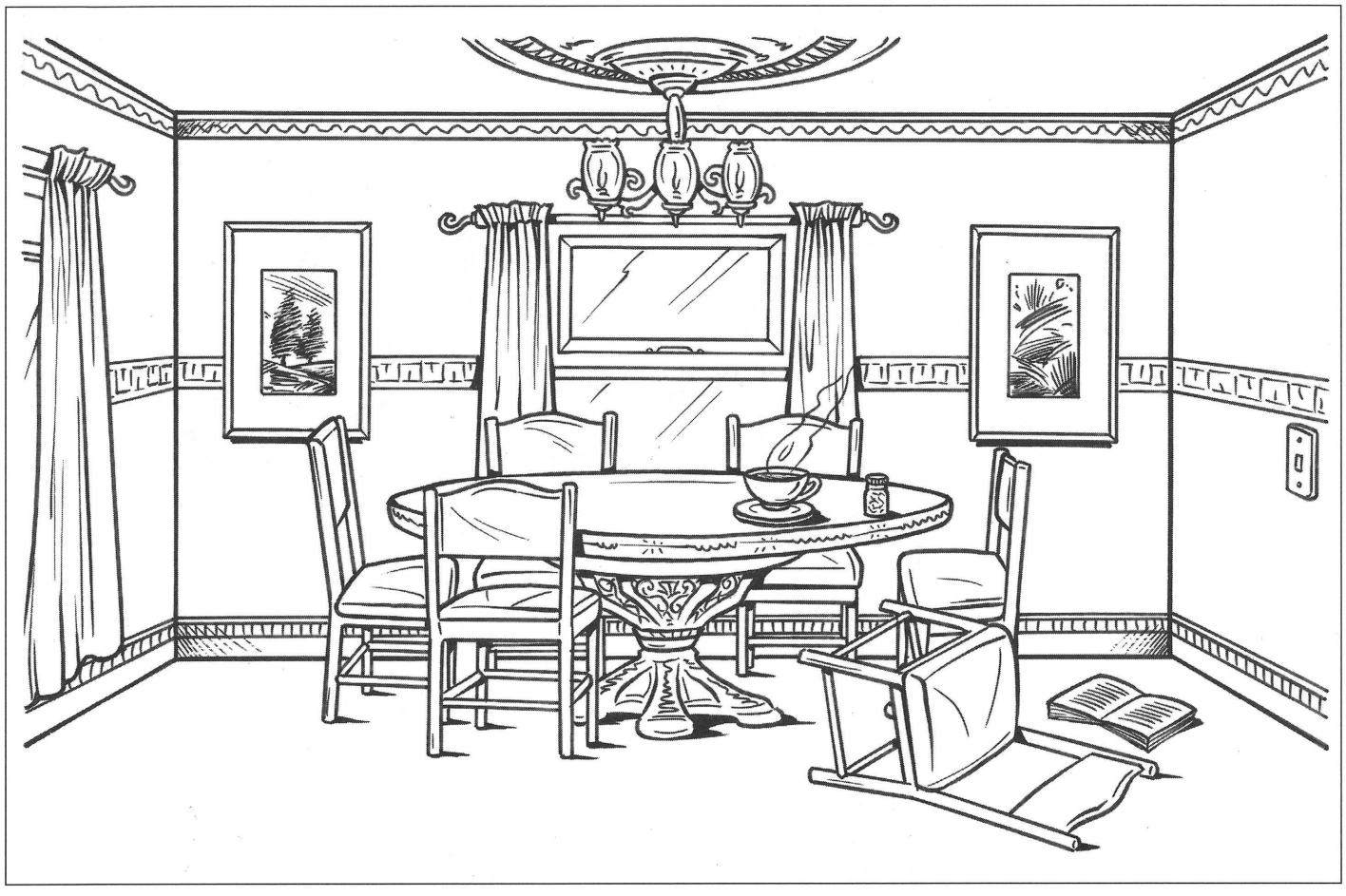
ISBN 978 1 92561 088 8

More information obtained by late morning

While the detectives were interviewing Mrs Hill and Patrick Sullivan, the crime scene examiners were completing their search of the Hill house and garden. A bottle of Barnaby's heart tablets was found on the dining table. Before she left, Mrs Hill showed the police officers where her husband usually kept them, in a kitchen cabinet with sliding glass doors. This is the same place in which the suicide note was found. Some fingerprints were found on the glass doors of the cabinet. They were developed and sent to the lab.

An almost full, cold cup of tea was on the kitchen table next to the tablets. An opened book was found on the floor, lying next to an overturned chair. Mrs Hill suggested that Barnaby may have used the cup of tea to help swallow the tablets. 'He often got into his pyjamas then came downstairs to take his tablets and read for a little while if he couldn't sleep and didn't want to disturb me,' she told the detectives.

Artist's sketch of dining room



As the investigators were conducting their interviews, the forensic scientists started sorting out the evidence that had been transported to the forensic laboratory, and Dr Simpson was preparing Barnaby Hill's body for the post mortem examination. While removing Barnaby's clothes, Dr Simpson found a note in his pyjama pocket (see below).

I think he knows about us. Be careful. He will try to hurt you. He is dangerous.

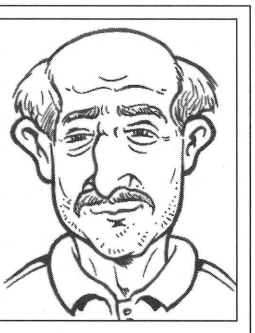
© Blake Education 2006

ISBN 978 1 92561 088 8

Information obtained by late afternoon .. PART 4

Use the information in this part to fill in Part 4 of your report.

POST MORTEM REPORT



Crime Reference Number 231078

Examination by DR S. J. SIMPSON

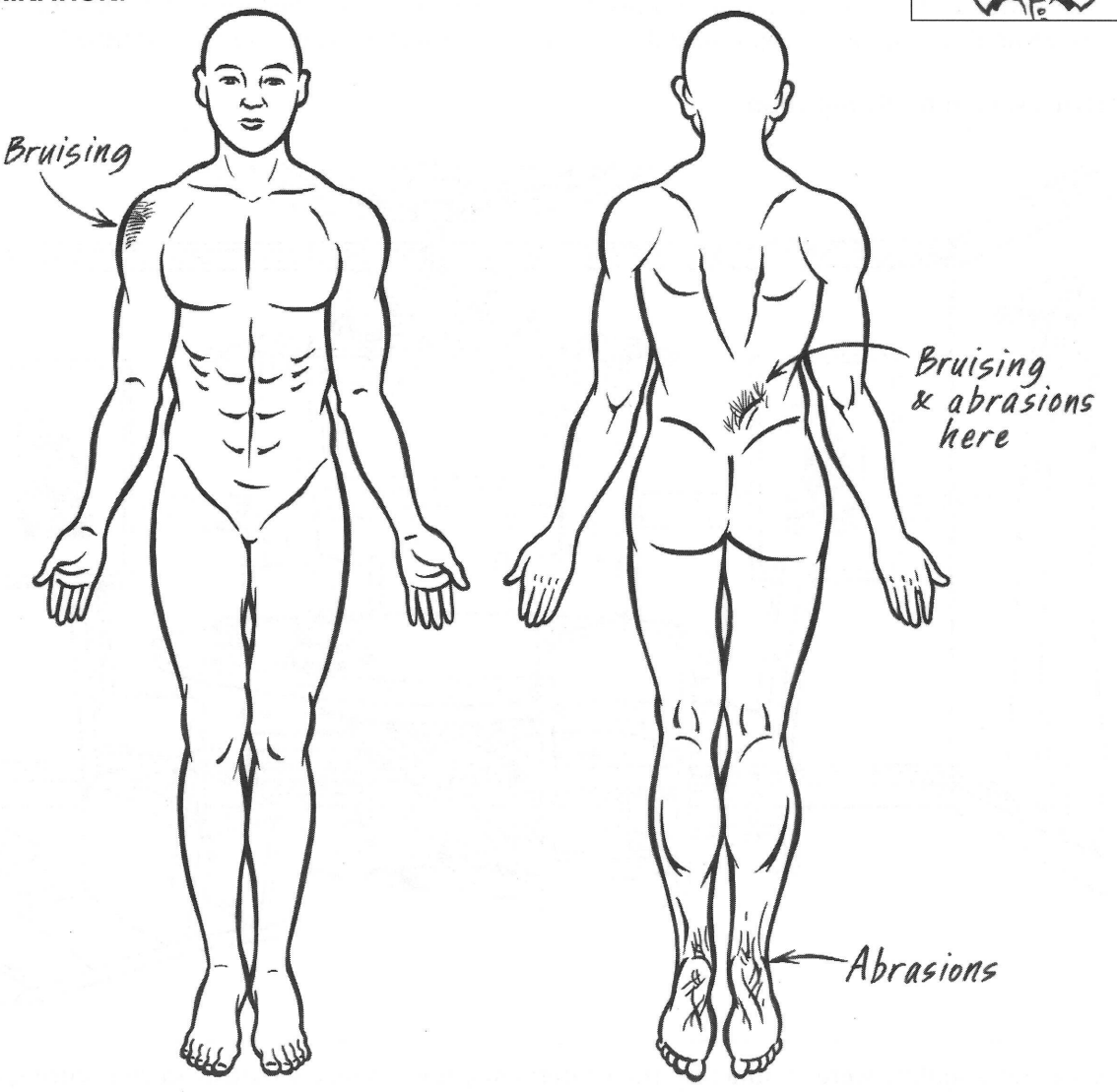
Name of Deceased BARNABY HILL

Age 60

Location 6 TELOPEA PLACE, BENWARRA

Offence SUICIDE OR MURDER

EXTERNAL EXAMINATION:



Body of a well-nourished, middle-aged Caucasian male. Residual weight 93 kg, Height 182 cm. There are no obvious injuries to the head, neck, face or mouth. Small bruises and abrasions to the shoulder region having occurred before death. Pupils are fixed and dilated. The sclera and conjunctiva are unremarkable. There are no obvious injuries to the chest, abdomen, or pelvis. Bruising and small abrasions are noted to the lumbar region (5 x 4 cm). Injuries are consistent with a fall from a chair, having occurred before death. Abrasions are noted overlying the posterior aspect of both heels. These injuries are again consistent with having occurred before death, possibly when unconscious and dragged over a rough surface.

INTERNAL EXAMINATION: Examination of the heart indicated evidence of angina for which medication was being taken.

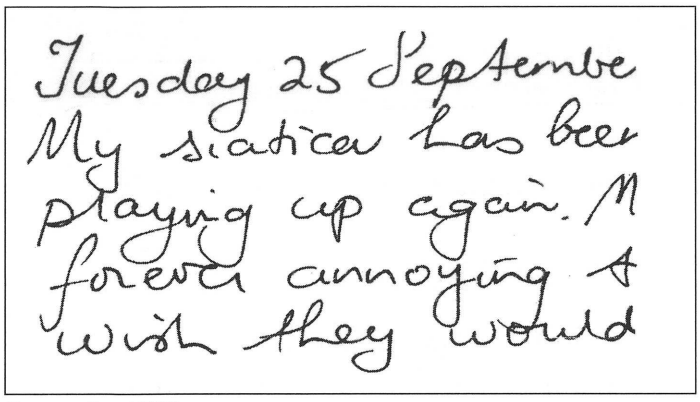
TOXICOLOGICAL ANALYSIS: A large amount of barbiturates and paracetamol detected, estimated to have been taken about an hour before death. Also alcohol, equivalent to 2 glasses of wine, approximately 2 hours before death. Traces of carbon monoxide consistent with the deceased being unconscious and breathing slowly whilst inhaling the fumes. Death occurred about 10 minutes after exhaust fumes were first inhaled.

ESTIMATED TIME OF DEATH: Between 11.30 pm and midnight

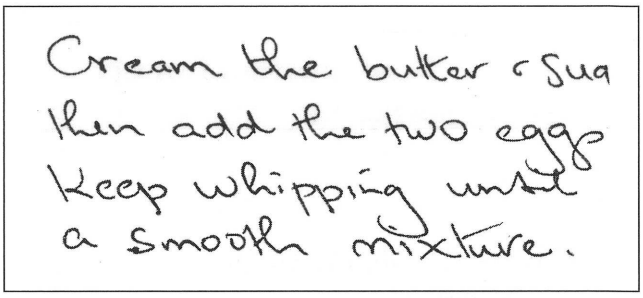
© Blake Education 2006

ISBN 978 1 92561 088 8

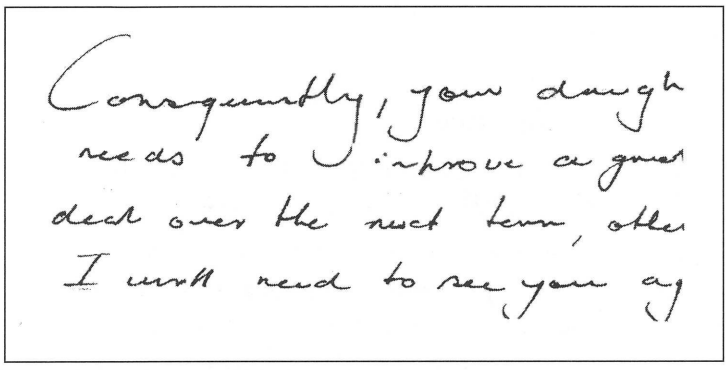
Handwriting samples


Tuesday 25 Septembe
My sciatica has been
playing up again. M
forever annoying &
wish they would

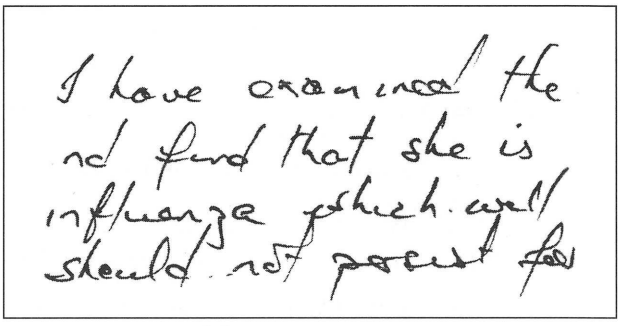
Barnaby Hill


Cream the butter & Sug
then add the two eggs
Keep whipping until
a smooth mixture.

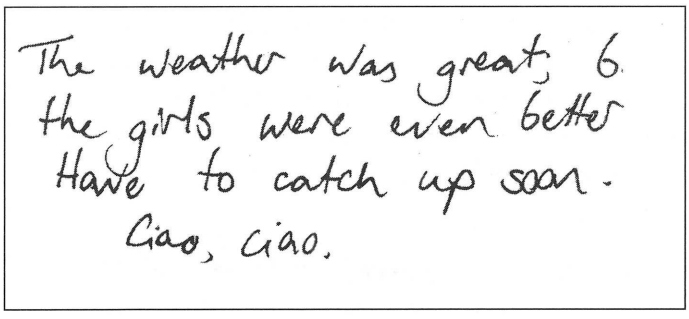
Helena Hill


Consequently, your daugh
needs to improve a great
deal over the next term, other
I will need to see you ag

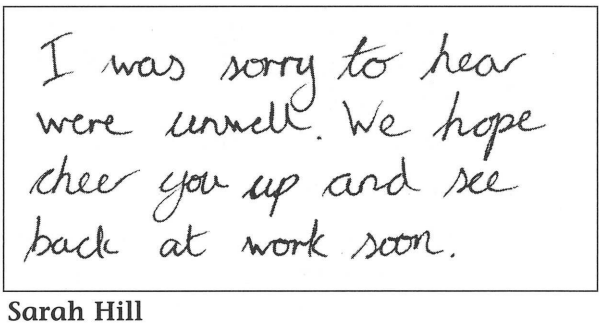
Isabella McDonald


I have examined the
rd and find that she is
influenza which will
should not present for

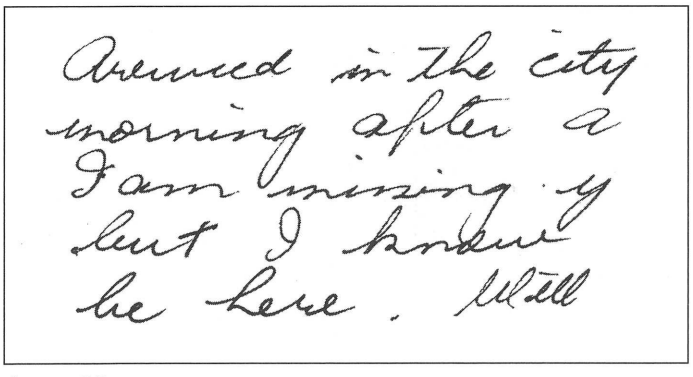
Brian McDonald


The weather was great; 6.
the girls were even better
Have to catch up soon.
Ciao, ciao.

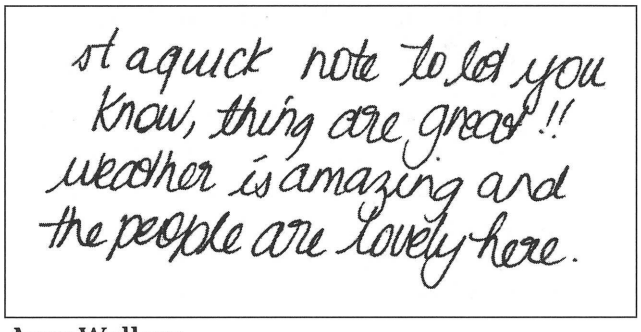
Timothy Hill


I was sorry to hear
were unwell. We hope
cheer you up and see
back at work soon.

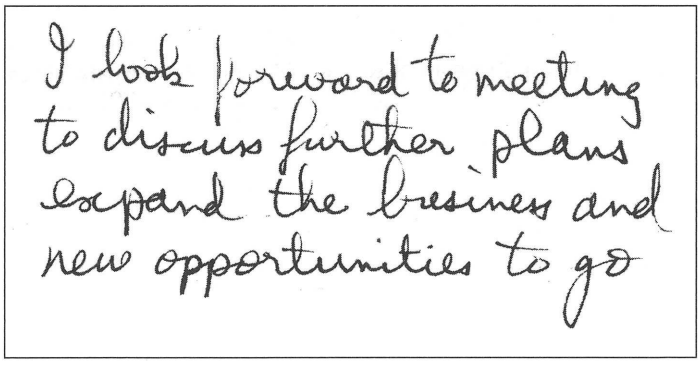
Sarah Hill


Arrived in the city
morning after a
I am missing my
but I know
be here. Will

Steve Nara


st a quick note to let you
know, thing are great!!
weather is amazing and
the people are lovely here.

Amy Wallace


I look forward to meeting
to discuss further plans
expand the business and
new opportunities to go

Lars Vogel

© Blake Education 2006

ISBN 978 1 92561 088 8

POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Timothy Hill

Taken by D.S. Davis Date 22.05.10 Time 10.00 am

Q Thanks Timothy for coming in to help us with our enquiries into what happened to your father.

Of course. It's a terrible shock. One minute he's with us, healthy and full of life, then he's gone. Just like that. I can't believe he killed himself. It's just not like him. Dad's a fighter, he wouldn't take his own life. I'm sure he was murdered, but then that's just as unbelievable.

Q Thanks, Timothy. We haven't assumed anything yet. We're keeping an open mind and following up every lead. Now, if you wouldn't mind, for the benefit of the tape, please state your full name, age, occupation, address and relationship to Barnaby Hill.

Sure. My name is Timothy John Hill. I am 24, and live in unit 7, 52 Grevillea Gardens in Benwarra. I'm a fitness instructor. Barnaby Hill is, or was, my father.

Q Thank you. Now I believe you were at your father's house on the evening before he died. Could you please tell us what you remember of the evening, starting from when you arrived.

Yes, well Amy, my girlfriend, and I arrived at 7 pm, because Mum commented how we were exactly on time. She complained about the McDonalds arriving too early. Dad was entertaining the McDonalds in the den, and Mum said dinner was ready as we arrived. My sister Sarah was there as well. It was a pleasant meal for most of the time. Mum and Dad don't really like Amy that much, but during the meal they were polite.

Q Now I believe there was a bit of an argument after Mr Nara arrived?

Yes, well there was some emotional discussion afterwards, which thankfully took the attention off Amy and me. Dad came back in after speaking to him. He boasted that Mr Nara was going to try to take him to court to get the money for the fence. Mum was not happy about it at all, but the McDonalds both agreed with Dad. Sarah took Mum's side, but Amy and I just stayed out of it. At the moment, if I say anything, Dad will probably take the opposite view and shout me down.

Q So you and your father haven't been getting along lately?

You could say that. First because I didn't follow his expectations and become a teacher, and now because of Amy. In fact, we've decided to get engaged. We told Amy's parents on Friday. I rang Mum, who has mixed feelings, and we decided to tell Dad after dinner last night.

Q So is this what you talked about after the McDonalds left? What time did you leave?

Yeah, Mum and Sarah said they'd do the dishes so Amy and I could talk to Dad. We only stayed about 20 minutes or so because he was already in a bad mood. So we told him and then left fairly soon after that.

Q Thank you. Now, can you tell us what was discussed after Mr Nara left?

Well I think they argued for the whole time we were eating dessert. First, it was about paying for the fence, but then Dr McDonald had a go at his wife for paying for her last haircut that she wasn't happy with. He said, pretty scornfully actually, that women always let themselves be taken advantage of by unscrupulous trades people. I was surprised when Dad took Mrs McDonald's side and suggested that Dr McDonald was being unfair in this case. I think Mum was quite happy to escape and carry the dessert dishes out and then get the coffee.

© Blake Education 2006
ISBN 978 1 92561 088 8

Q So while your mother went out to the kitchen to get the coffee, what did everyone else do?

Well it was strange really. Dr McDonald reminded his wife that he had to check up on a patient in hospital, then he got up and said he'd help Mum get the coffee before either Sarah or I could offer. I think Mum was a bit upset over all the arguing, but Dr McDonald has never been the caring sort, and certainly never towards Mum. He and his wife are more Dad's friends. In fact, I think Dad and Isabella were an item once, before Dr McDonald came on the scene. Mrs McDonald seemed surprised that he was going to the hospital that evening. She said that the doctor hadn't mentioned it to her before.

Q So while Dr McDonald joined your mother in the kitchen, what was discussed at the table?

I think Dad deliberately changed the subject. He asked Mrs McDonald about her recent holiday in Europe. She was suddenly all coy, stroking his arm and batting her eyelashes as she talked about her trip. The rest of us were just ignored. Sarah whispered to Amy and me, asking if we'd told Dad yet – about our engagement, that is. We shook our heads. It obviously wasn't a good time, but we couldn't leave it any longer because too many other people already knew.

Q Yes, I see. How soon did the McDonalds leave after finishing their coffee?

I'd say around about 9.30 I think. Dr McDonald seemed to be in a bit of a hurry, and when Dad commented on this, Dr McDonald said that he'd remembered that he had to collect something from home first before visiting the patient, and he didn't want to arrive too late at the hospital.

Q So what did your father do after the McDonalds left?

Well, Mum wasn't feeling well. She had a headache, which wasn't surprising considering the way Dad spoke to her about the fence. Sarah and I told her to go to bed and we'd see to the dishes. In the end, Sarah helped Mum clear the dishes, and Amy and I went into the den with Dad.

Q So how did your father react to the news of your engagement?

Not well, as expected. He was angry and very rude to Amy. He threatened that he would change his will and leave everything to Sarah if Amy and I got married.

Q So the meeting didn't go well?

It did, in the sense that it cleared the air and brought everything out into the open. Of course everything has changed now because he's dead. He was a difficult man and not easy to love, but he didn't deserve to die that way. I hope you get the person responsible.

Q Yes, well we have a large team working on that. Now, you left your father about 10 pm?

Yeah, that'd be about right, because we were home by 10.30.

Q Timothy, can you think of anyone who might want to murder your father? Any threats?

Not that I'm aware of. He's rubbed a few people up the wrong way over the years. Steve Nara is obviously the most recent, and as you probably know, Dad's been accused of molesting a man called Lars Vogel when he was a boy in Dad's class over 20 years ago. But that doesn't mean they would murder him.

Q No, that's right. Now one more thing. This suicide note was found at your parents' house this morning. Do you recognise the writing? [Tim was shown a copy of the suicide note.]

Well, I have to say that it certainly looks like Dad's handwriting. I know Mum thought that too when she was shown the note. But Dad's just not the type to do that, to take his own life.

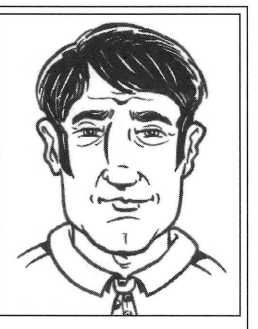
Q No, well we're keeping an open mind on that at this stage. Thank you Timothy for coming in. If we have any further questions we'll be in touch.

This statement of 2 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed T Hill

© Blake Education 2006

ISBN 978 1 92561 088 8



POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Brian McDonald

Taken by D.S. Davis Date 22.05.10 Time 11.00 am

Q *Thank you for coming in Dr McDonald. As you know, we'd like to ask you some questions regarding your friendship with Barnaby Hill, and what happened last night. Could you start by giving your full name, age, occupation and address. Also your spouse's name and your relationship to the deceased.*

My name is Brian Michael McDonald. I am a doctor, and aged 58. I live at 73 Cherrytree Lane, Benwarra Heights. My wife is Isabella Anna McDonald and Barnaby Hill is an old friend from our school days.

Q *So you knew Barnaby Hill very well. Could you please look at this suicide note and tell us if you recognise the writing? [Dr McDonald was shown the suicide note.]*

The writing very definitely looks like Barnaby's handwriting.

Q *Thank you, Doctor. Now yesterday evening you joined the Hill family for a meal. Can you tell us when you arrived, and take us through what happened during the evening?*

Of course, Sergeant. Isabella and I arrived just after 6.30. Helena was in a bit of a flap, so Barnaby took us into the den for some drinks where we stayed until after young Tim and his girlfriend arrived. After some small talk, we moved into the dining room and had a very nice meal. Helena Hill is a very good cook. Isabella and I left about 9.30 because I had promised to check on a patient before going home to bed.

Q *Can you tell us how Barnaby was affected by Steve Nara's visit?*

Well naturally Barnaby was very annoyed that the man turned up on his doorstep on a Saturday night, asking for money. Barnaby told him to get lost. If there's a dispute over payment for a job, then Saturday night, at the customer's house, is not the place to sort it out.

Q *No. Now can you tell us if any threats were made?*

No, I don't think so, only that the man was going to see Barnaby in court.

Q *Thank you. Now you say you left at about 9.30 pm. Can you tell us what you did after that?*

Isabella and I drove home together. I picked up some items that I needed, then drove to the hospital to visit Mrs Pagonas who's just had a hip replacement. I spoke to the nurse on duty then checked on my patient. Unfortunately she was asleep, so as I was quite wide awake and Isabella had gone to bed, I decided to catch up on some bookwork in my office before going home.

Q *Can you tell us when you left the hospital, and who can verify that?*

I left fairly late, about midnight I think. I'm not exactly sure, but I did see the nurse again. I also filled out the staff sign-in log book, so that should record exactly when I arrived and left that evening.

Q *Thank you Doctor, that's all for the moment. We appreciate your cooperation on this.*

That's fine. I'd just like to see that whoever did this is caught and gets what they deserve.

This statement of 1 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed B M Brown

© Blake Education 2006
ISBN 978 1 92561 088 8

POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Sarah Hill

Taken by P.C. Collins Date 22.05.10 Time 11.00 am

Q Thank you Sarah for helping us with our enquiries into the death of your father. Could you, for the benefit of the tape, state your full name, age, occupation, address and relationship to the deceased.

My name is Sarah Zoe Hill, I am 22 and live in unit 14, 47 Percy Place, Benwarra. I'm a student.

Q Thank you Sarah. May I ask, is that your natural hair colour? It's a lovely brown.

Well yes, but I can't see what that's got to do with finding out who killed my father.

Q Can you tell me about the afternoon and evening before your father died? When did you arrive at your parents' house?

I arrived during the afternoon, about 4 pm. I'd borrowed Dad's car for a couple of days while mine was having some work done by a friend. My car was ready, so I drove to my parents' house and then Dad drove me over to pick up my own car. Mum asked me for dinner, so I followed Dad back home.

Q So you arrived back about what time? And what did you do then?

Dad and I got back before 5 o'clock. We watched the early news because we knew we wouldn't have time later on, and then I helped Mum finish getting dinner.

Q Now I believe there was some conflict after the arrival of the fencing contactor, Steve Nara?

Only that Dad started to get arrogant and aggressive about not paying for the fence because it's not on the correct boundary. Dr McDonald agreed with him, so that really added fuel to the fire.

Q I was told there was once a relationship between your father and Mrs McDonald?

Oh that was before Mum was on the scene. Mrs McDonald was a teacher at the same school as Dad. Later he became principal and she was one of his staff, so they've known each other for a long time.

Q But there is nothing between them now?

Only a friendship I think. They seem close friends, but that's to be expected I suppose, since they've known each other that long.

Q So there was nothing strange in your father taking Mrs McDonald's side when Dr McDonald was criticising her about paying a hairdresser's bill? It didn't upset your mother?

Mum's never really liked the McDonalds. Isabella thinks she is better than Mum and that Dad settled for second best by choosing Mum over her. Dad just laughs and says that Mum's jealous that he can still attract another woman. I think he deliberately encourages Mrs McDonald just to tease Mum and the doctor, and maybe to boost his ego.

Q Okay, thanks Sarah. Now what time did you leave?

Just before 10 I think. I helped Mum clear the coffee cups and stack the dishwasher, and then I left. Tim and Amy were still with Dad. Dad's voice was pretty loud, and I could hear him threatening to cut Tim out of his will.

Q Thank you Sarah, I think that's all for now. We'll let you know of any new developments. Please let us know if you remember anything else that you think might be helpful.

This statement of 1 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed Sarah Hill

© Blake Education 2006

ISBN 978 1 92561 088 8

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.
Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name BARNABY HILL

Address 6 TELOPEA PLACE, BENWARRA

Offence SUICIDE OR MURDER

Location 6 TELOPEA PLACE, BENWARRA

Taken by SIMPSON

Rank PATHOLOGIST

Date 22.05.10

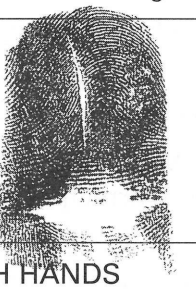
Crime Reference Number 231078

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

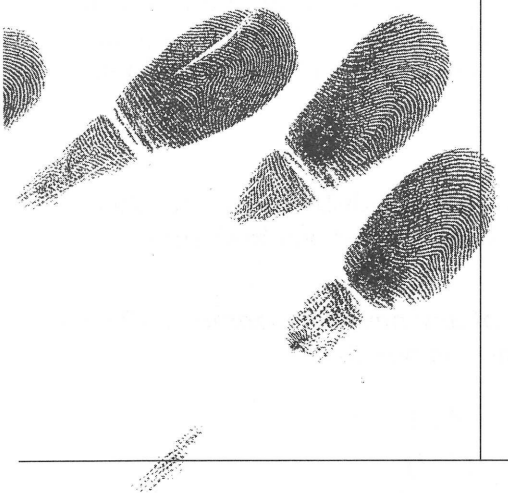
ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

© Blake Education 2006

ISBN 978 1 92561 088 8

BARNABY HILL
Right Hand

Left Hand

© Blake Education 2006

ISBN 978 1 92561 088 8

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.
Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name HELENA HILL

Address 6 TELOPEA PLACE, BENWARRA

Offence MURDER

Location 6 TELOPEA PLACE, BENWARRA

Taken by COLLINS

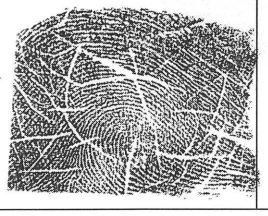
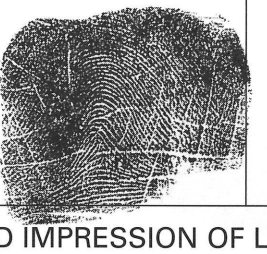
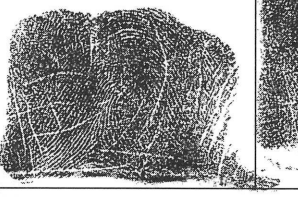
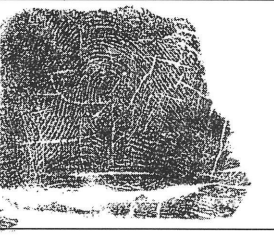
Rank P.C.

Date 22.05.10

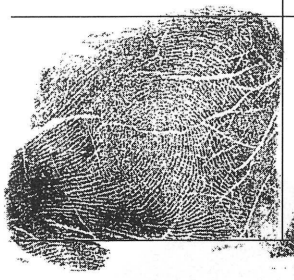
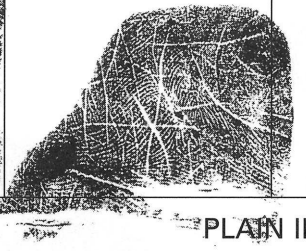
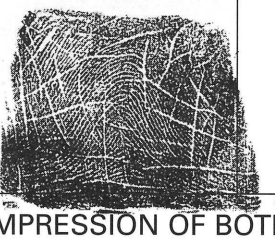
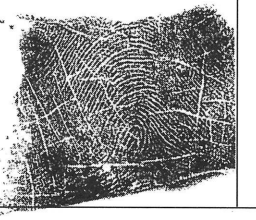
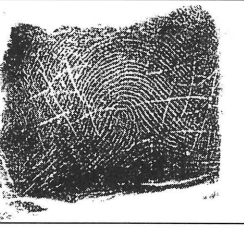
Crime Reference Number 231078

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

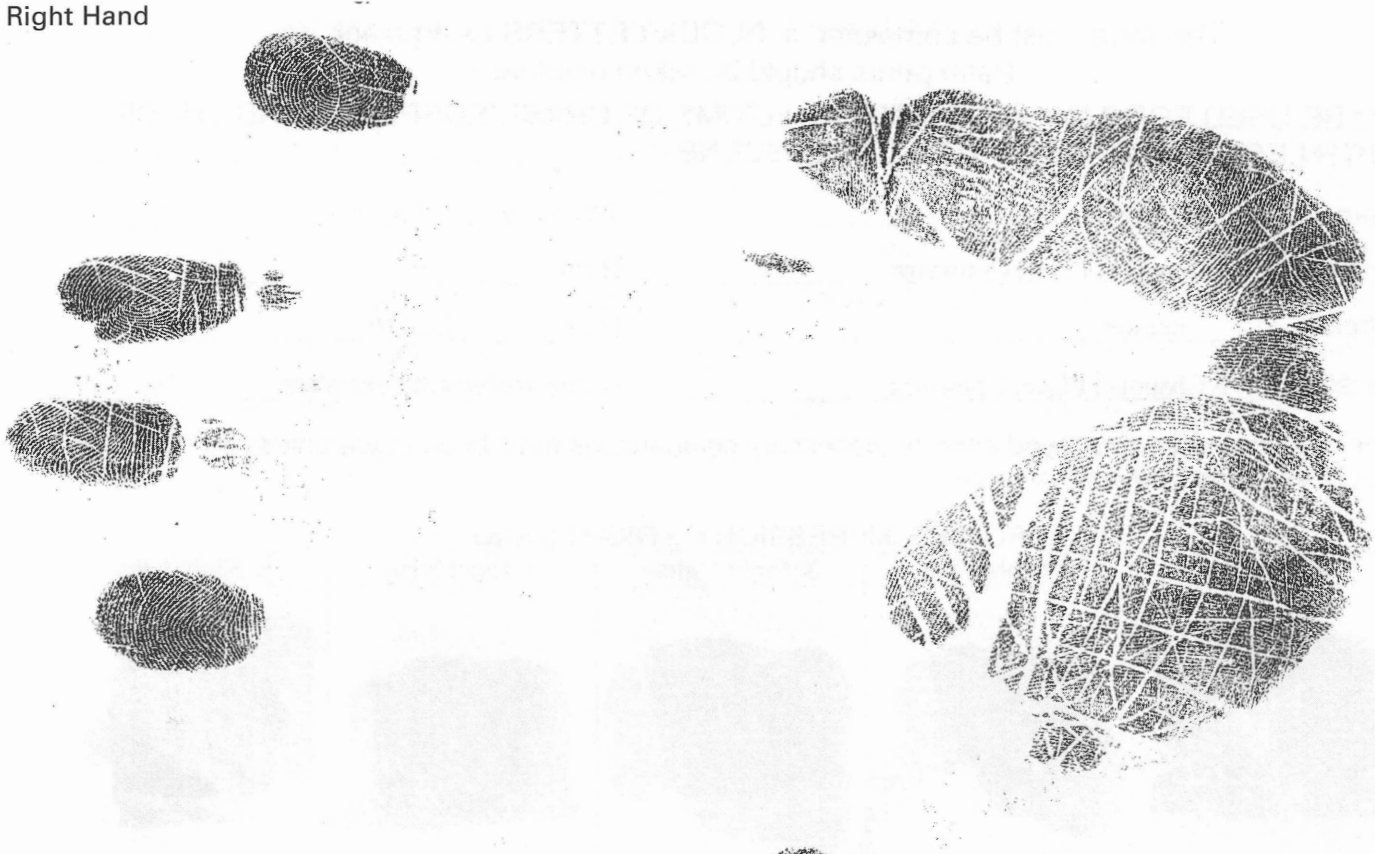
PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together

© Blake Education 2006

ISBN 978 1 92561 088 8

HELENA HILL
Right Hand



Left Hand

ISBN 978 1 92561 088 8

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.
Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name DR BRIAN MCDONALD

Taken by FLETCHER

Address 73 CHERRYTREE LANE, BENWARRA HEIGHTS

Rank P.C.

Offence MURDER

Date 22.05.10

Location 6 TELOPEA PLACE, BENWARRA

Crime Reference Number 231078

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

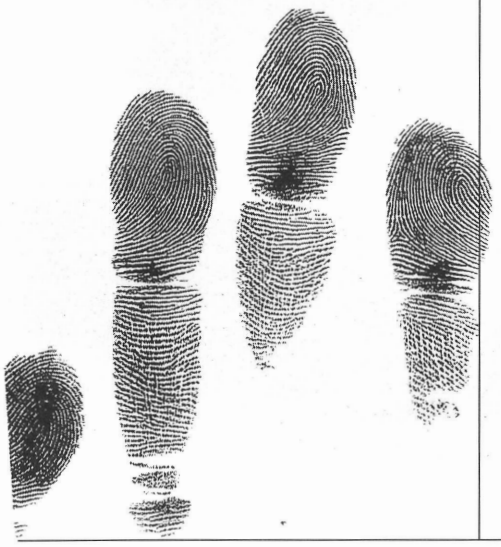
ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little

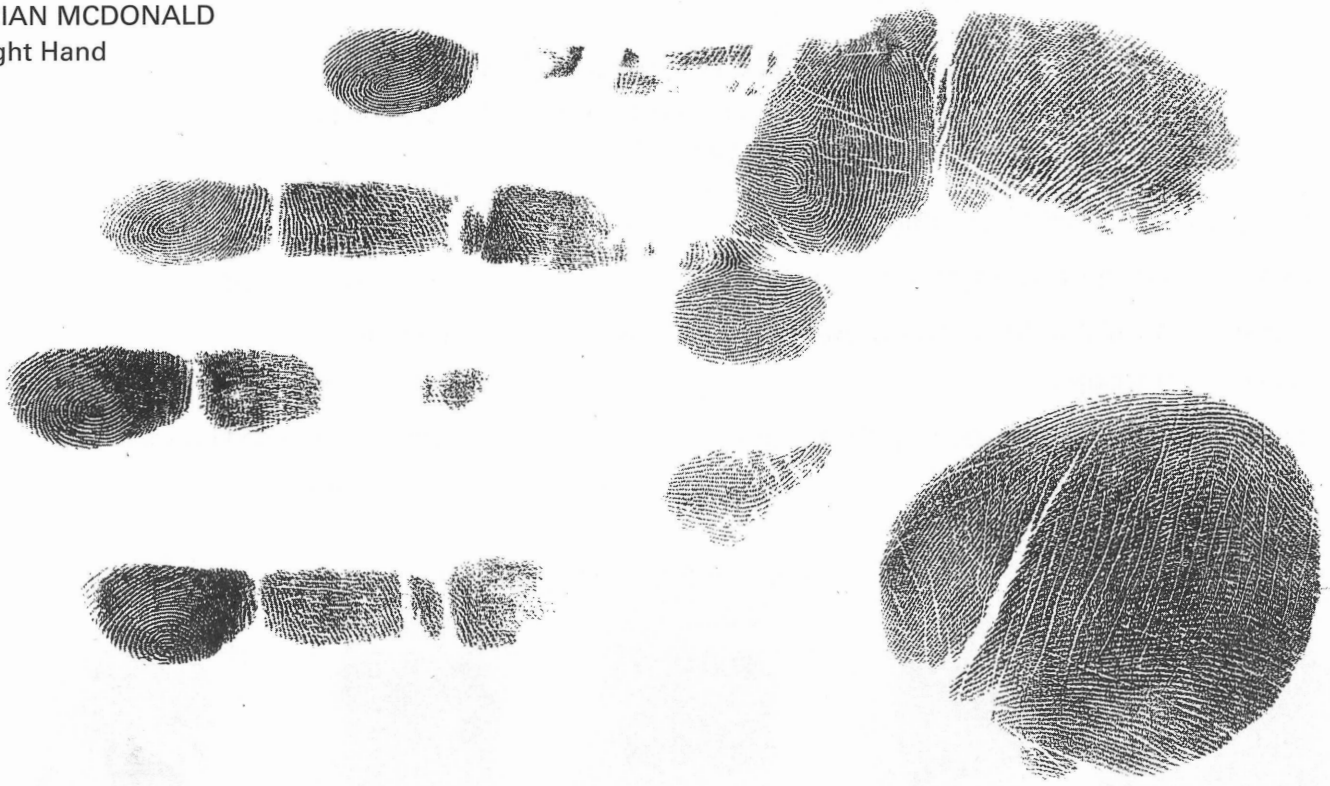
PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

© Blake Education 2006

ISBN 978 1 92561 088 8

BRIAN MCDONALD
Right Hand



Left Hand

© Blake Education 2006

ISBN 978 1 92561 088 8

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.
Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name ISABELLA MCDONALD

Address 73 CHERRYTREE LANE, BENWARRA HEIGHTS

Offence MURDER

Location 6 TELOPEA PLACE, BENWARRA

Taken by FLETCHER

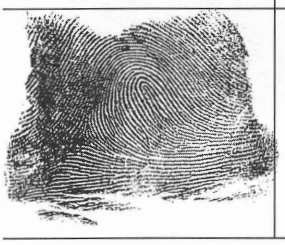
Rank P.C.

Date 22.05.10

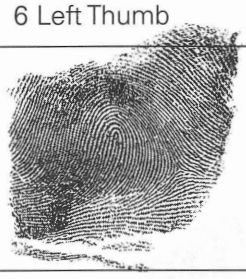
Crime Reference Number 231078

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

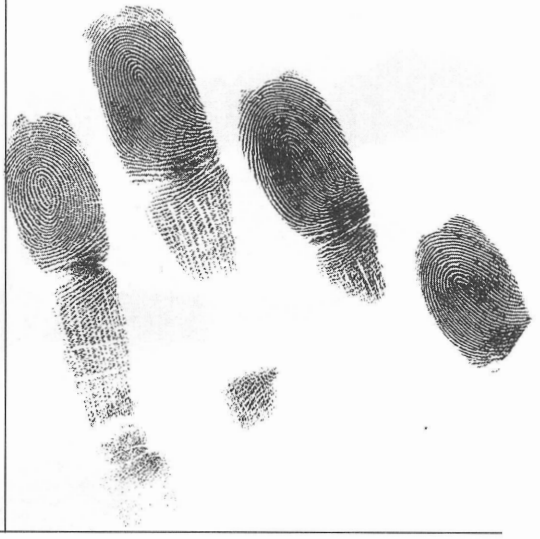
ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

© Blake Education 2006
ISBN 978 1 92561 088 8

ISABELLA MCDONALD
Right Hand

Left Hand

© Blake Education 2006

ISBN 978 1 92561 088 8

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.
Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name TIMOTHY HILL

Address 7/52 GREVILLEA GARDENS, BENWARRA

Offence MURDER

Location 6 TELOPEA PLACE, BENWARRA

Taken by MATTHEWS

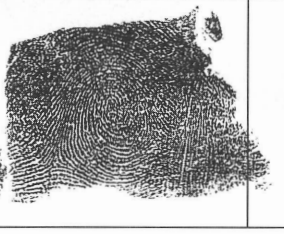
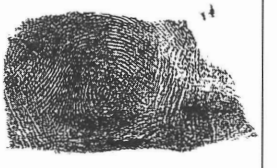
Rank D.I.

Date 22.05.10

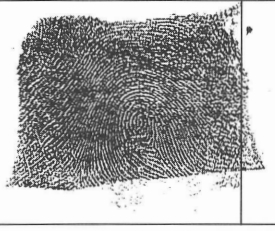
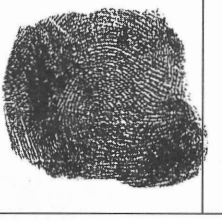
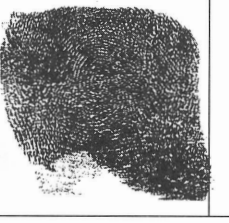
Crime Reference Number 231078

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

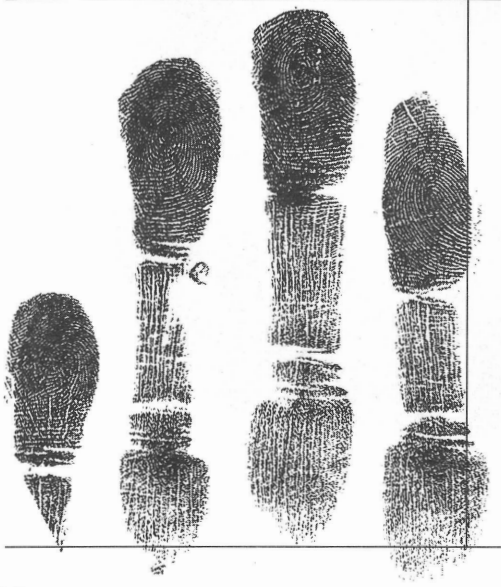
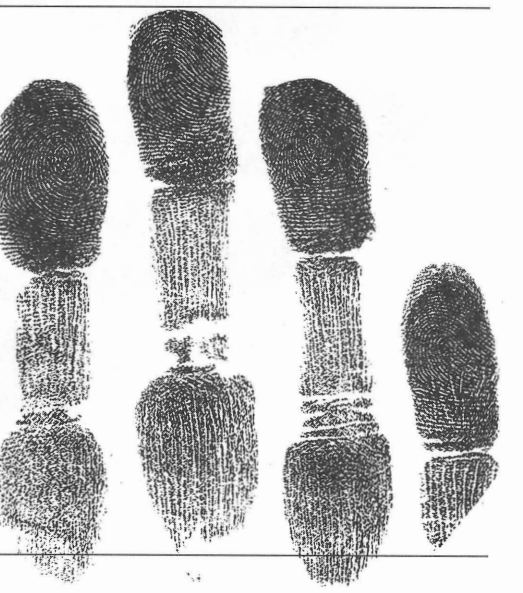
ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little
				

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little
				

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

© Blake Education 2006
ISBN 978 1 92561 088 8

TIMOTHY HILL
Right Hand

Left Hand

© Blake Education 2006

ISBN 978 1 92561 088 8

ELIMINATION FINGERPRINT FORM

This form must be completed in BLOCK LETTERS using black ink.
Palm prints should be taken overleaf.

TO BE USED FOR TAKING PRINTS OF VICTIMS OF CRIME, SUSPECTS AND THOSE WITH LEGITIMATE ACCESS TO A CRIME SCENE.

Name SARAH HILL

Address 14/47 PERCY PLACE, BENWARRA

Offence MURDER

Location 6 TELOPEA PLACE, BENWARRA

Taken by COLLINS

Rank P.C.

Date 22.05.10

Crime Reference Number 231078

These prints will be destroyed after the necessary comparisons have been made unless the donor has been found guilty of the crime.

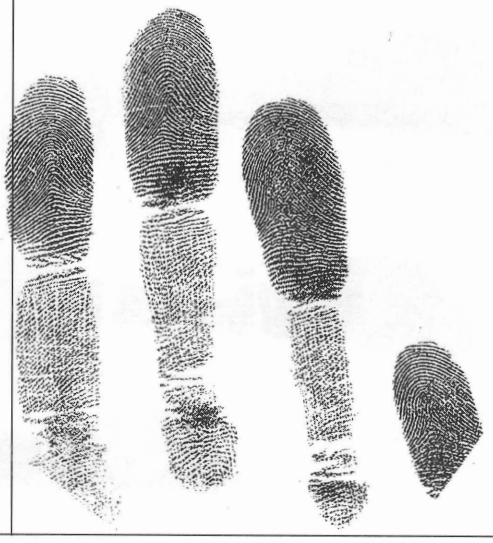
ROLLED IMPRESSION OF RIGHT HAND

1 Right Thumb	2 Right Fore	3 Right Middle	4 Right Ring	5 Right Little

ROLLED IMPRESSION OF LEFT HAND

6 Left Thumb	7 Left Fore	8 Left Middle	9 Left Ring	10 Left Little

PLAIN IMPRESSION OF BOTH HANDS

LEFT HAND Four fingers taken together	TWO THUMBS Taken together LEFT RIGHT	RIGHT HAND Four fingers taken together
		

© Blake Education 2006
ISBN 978 1 92561 088 8

SARAH HILL
Right Hand

Left Hand

© Blake Education 2006

ISBN 978 1 92561 088 8

Some practical activities PART 5

Use the information in this part to fill in Part 5 of your report. You will need the physical evidence to analyse.

1 Using chromatography to compare the inks in the pens

Both the suicide note and the note found in Barnaby's pocket were written in black, with water-soluble, felt-tipped pens. One such pen was found on the bench in Helena Hill's kitchen. The police also visited the suspects' homes and found four more similar pens that may have been used to write either or both notes. The owners of these pens were Isabella McDonald, Sarah Hill, Timothy Hill and Steve Nara.

Your task is to use paper chromatography to separate the dyes used in the pens, to see which pen may have been used to write the suicide note.

What to do:

1 Take a piece of chromatography or filter paper and cut it into seven strips about 1.5 cm wide and 10 cm long.

2 Use a pencil to rule a line 1 cm from the bottom of each strip. At the top of each strip, write the initials of one of the pen's owners or CSS for 'crime scene suicide note' and CSW for 'crime scene warning note'.

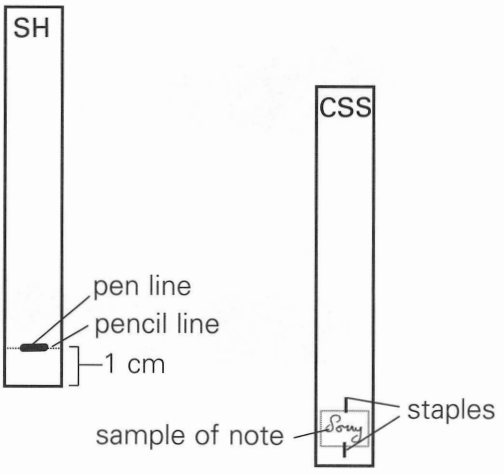
3 Using each of the five pens, mark a small line on the corresponding strip, at the pencil line height as shown at right.

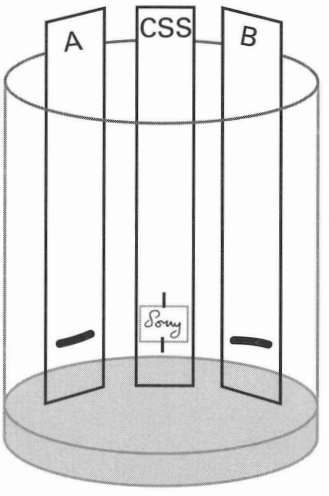
4 Cut a small piece of the suicide note leaving some white unmarked paper above and below the writing. Use two staples, above and below the writing, to attach this ink to the ruled line on the CSS strip. Repeat with the warning note for the CSW strip.

5 Take a 250 mL beaker and place 0.5 cm of water in the bottom. Stand the strips in this beaker so that all of the ink marks **are above** the surface of the liquid.

6 Watch the water soak up the paper and carry some of the dyes in the black ink with it. Remove the strips from the beaker before the colours reach the top.

7 The pen that gives the same pattern of dyes (called a chromatogram) as the pattern from the ink used in the suicide note is probably the same pen that was used to write the note, and similarly with the warning note.





2 Comparing fibres

Some dark blue fibres were found caught in the cloth tied around the garden hose that was attached to the car exhaust pipe. They are thought to be woollen fibres that may have come from the clothes that the murderer was wearing while attaching the hose to the exhaust pipe.

These people all own dark blue knitted garments: Dr Brian McDonald, Timothy Hill, Helena Hill and Steve Nara. Fibres were collected from each garment and provided for your inspection.

Use a microscope or magnifying glass to compare the fibre samples collected with those found at the crime scene. It may also be useful to compare them under ultra-violet light if this is available.

3 Comparing hair

The long brown hairs found caught in the seatbelt webbing of the Volvo were collected. Hair samples were also collected from Sarah Hill, Helena Hill, Isabella McDonald and Amy Wallace because they all had hair that was at least 25 cm or longer. Use a magnifying glass to compare the sample collected from the crime scene with the hair strands taken from the four women. Record which hair colour and texture is the closest match to the strands found at the crime scene.

© Blake Education 2006

ISBN 978 1 92561 088 8

Information available early next day PART 6

Use the information in this part to fill in Part 6 of your report.

POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Isabella McDonald

Taken by P.C. Collins Date 22.05.10 Time 4.30 pm

Q *Thanks for coming in Mrs McDonald. Before we start, for the benefit of the tape, would you mind stating your full name, age and address. Also your occupation and your relationship to the deceased.*

Of course not, dear. My name is Isabella Anna McDonald. I am 55 and live at 73 Cherrytree Lane, Benwarra Heights. I am a teacher and Barnaby Hill was a friend of ours.

Q *Thank you, Mrs McDonald. Could you please tell us how long you have known Mr Hill?*

I first met Barnaby about 30 years ago, when we were both teaching at the same school.

Q *Barnaby was your boyfriend around that time, was he not?*

Yes, well we did go out together for several months before Barnaby started dating Helena and Brian and I got together.

Q *Did you also have a more recent affair with Barnaby Hill?*

Well, yes we did, but we were going to end it.

Q *So, had you ended the affair?*

Brian found out a while ago, and threatened to divorce me, which would have caused quite a scandal. So Barnaby and I agreed to cool it for a bit and just be friends. Unfortunately, last week I was comforting Barnaby about the problem with Lars Vogel, and Brian found out. Now of course, it's ended anyway.

Q *Mrs McDonald, can you tell me what you and your husband did after you left the Hill house on Saturday evening?*

Of course. Brian and I drove straight home. Brian had to collect some papers from his study to take to the hospital, so I went to bed with a book.

Q *Can you tell me the time that all this happened, please?*

We left the Hills' house about 9.30. I know that because Brian was making such a fuss about it. So we would've arrived home by 9.45. I think it was a few minutes before 10 o'clock when Brian left again.

Q *And what time did Brian return home? Were you still awake?*

No, I'm not sure. I read until about 10.30, but then turned the light off to get some sleep. I'm not sure exactly what time Brian got home, but it was probably before midnight because I got up to go to the bathroom just after 12.30 and Brian was in bed asleep. I expect he must have arrived home about 11, because the hospital is only about five kilometres from our house. I'm sorry, that's the best I can do. Over the years I've gotten used to not waiting up for him. A doctor's life is never predictable.

Q *Thank you, Mrs McDonald. We appreciate your help in this.*

Yes, well anything I can do to help. It's a dreadful business. Barnaby was a special, good friend who I've known for years. I don't believe he took his own life – it's just not like him. But he did get people offside at times. I just hope you get whoever did this.

This statement of 1 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed I A McDonald

© Blake Education 2006

ISBN 978 1 92561 088 8

POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Steve Nara

Taken by P.C. Collins Date 22.05.10 Time 3.30 pm

Q Thanks for coming in Mr Nara. For the benefit of the tape, could I just get you to start with your full name, age, occupation, address and your relationship to the deceased.

Okay then. My name is Steve Satu Nara. I am 32 years of age and live at 15 Solitaire Street, Erindale. I am a fencing contractor and I replaced an old fence for the Hills last month for which Mr Hill now refuses to pay me.

Q Thank you. So Mr Hill owed you money?

Yeah, about $3000. Hill claims that the fence is not on the correct boundary, even though I put it exactly where the old fence was. So he says it was my job to check the boundary. He's wrong. It's the owner's responsibility to get the land surveyed and to tell the contractor where to put the fence.

Q Hmm, I can understand how that could cause conflict. Now, can you tell me what happened when you called in at the Hills' house on Saturday evening please?

There's not much to tell. I wanted to try one more time to get my money before the hassle of taking him to court.

Q I believe you didn't get a very warm welcome. What happened?

No, Hill wouldn't listen. He pushed me off the doorstep so I almost fell in the garden beside the door. Then he closed the door in my face. I could charge him with assault, couldn't I?

Q Well yes, except he is no longer alive. Can you tell me what you did next?

Yeah, well I was so angry, I decided not to go home but went to the pub. It's the one on the crossroads near my house at Erindale. Anyway, I met Lars Vogel there, of all people. We've known each other casually for years, but it wasn't 'til Saturday night that I heard about how Hill molested boys like Lars. So we talked for a while about how people like Hill think they can do what they like and get away with it.

Q So what then?

Not much. Lars had a bit much to drink, so I asked him around to my place for coffee to sober up. I decided to show him the fancy house Hill lived in, so we drove past his house on the way home.

Q About what time was this?

About 10 I think. We had coffee, then about 11 o'clock Lars thought he was okay to drive home.

Q So did you go back to Telopea Place?

No, of course not. Lars walked over to the pub to get his car, and I hit the sack. The first I knew about what happened was on the news in the morning.

Q Well thank you Steve. That's all for now. We'll be in touch if we have any further questions.

This statement of 1 pages signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed S Nara

© Blake Education 2006

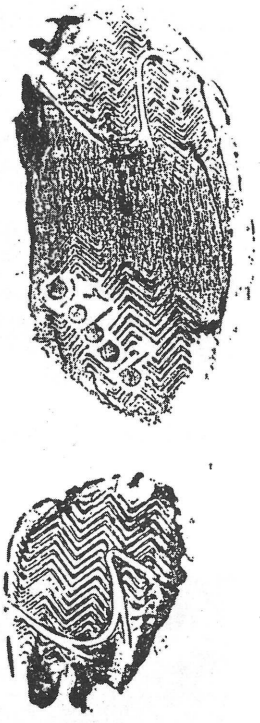
ISBN 978 1 92561 088 8

Prints of suspects' shoes (Scale: reduced to 30 per cent)

Helena Hill's gardening shoe

Sarah Hill's right sports shoe

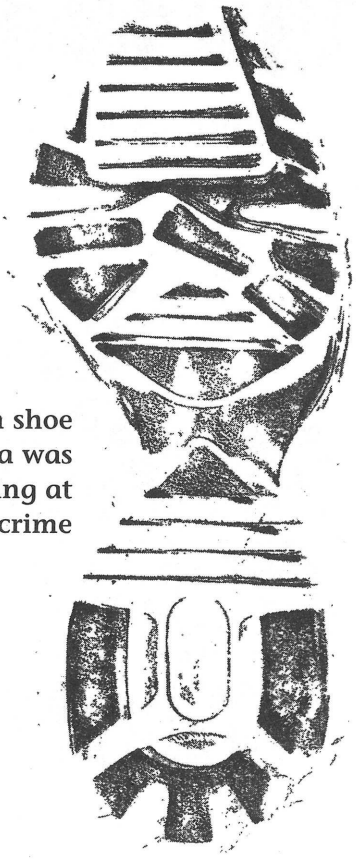
Dr Brian McDonald's right sports shoe



Amy Wallace's right sports shoe

Lars Vogel's right sports shoe

Timothy Hill's right sports shoe

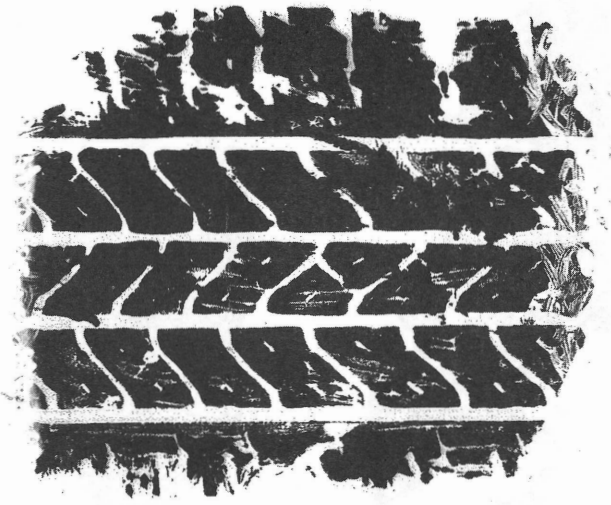


Print from shoe Steve Nara was wearing at time of crime

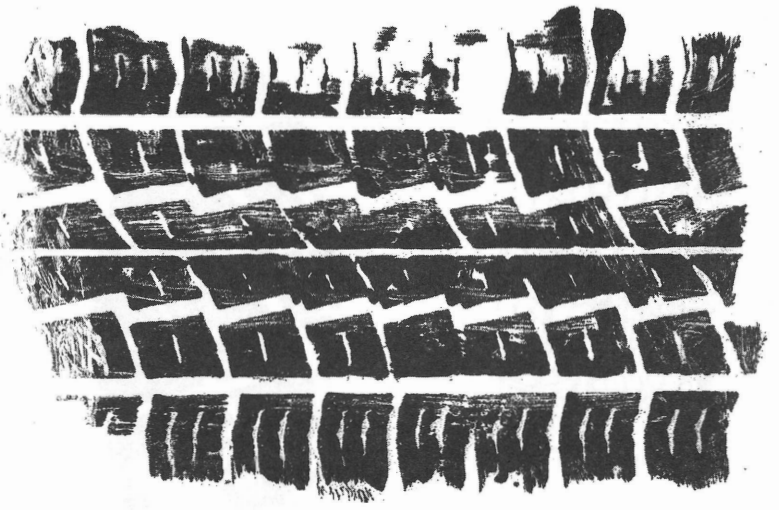
© Blake Education 2006

ISBN 978 1 92561 088 8

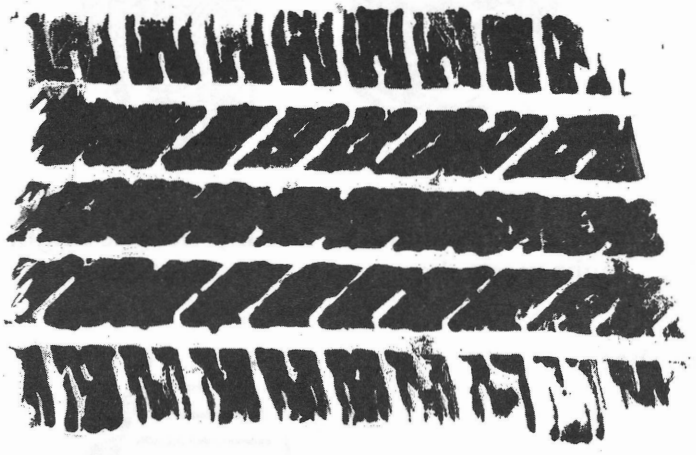
Suspects' car tyre prints (Scale: reduced to 40 per cent)



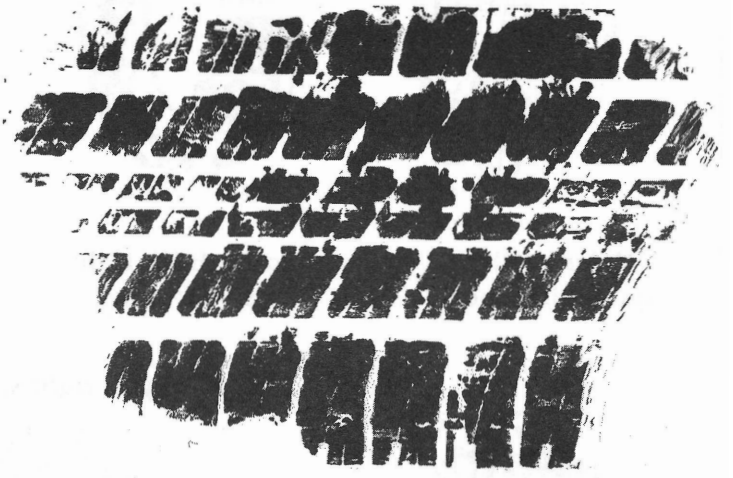
The Hills' Volvo (2002, white)



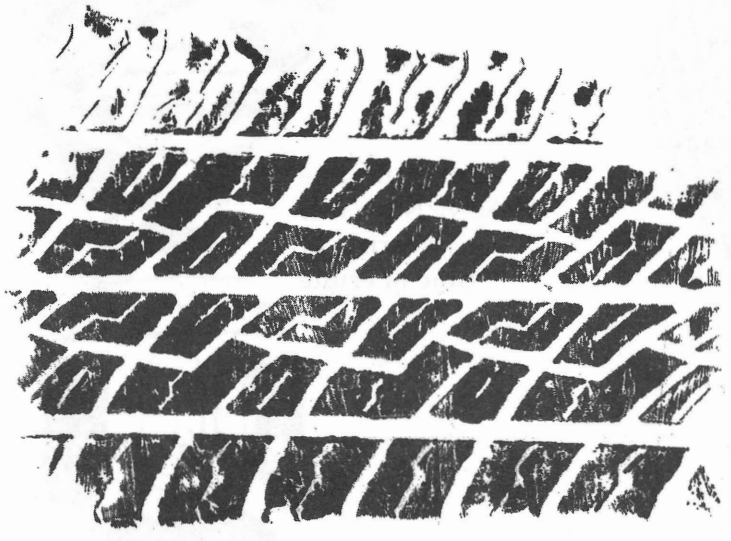
Sarah Hill's Ford Laser (1988, brown)



The McDonalds' BMW (2004, white)



Timothy Hill's Mazda MX6 (1995, red)



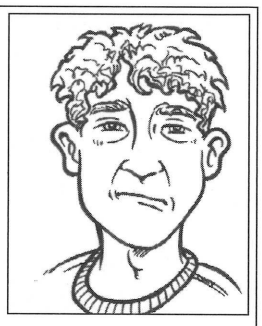
Steve Nara's Ford Ute (1984, cream)

Lars Vogel's Camira (1978, white)

© Blake Education 2005
ISBN 978 1 92561 088 8

Final evidence from a suspect and the lab .. ·· .. ·· .. PART 7

Use the information in this part to fill in Part 7 of your report.



POLICE CONSTABULARY
STATEMENT OF WITNESS

Statement of (name of witness) Lars Vogel

Taken by D.I. Brown Date 22.05.10 Time 5.30 pm

Q *Thank you Lars for coming in to help with our enquiries. For the benefit of the tape, could you please state your name, age, occupation, address and relationship to the deceased.*

My name is Lars Theodore Vogel. I live in unit 21, at 55 Nugent Avenue in Benwarra. I am 31 years of age and I was a student in Mr Hill's class about 20 years ago. At present I'm unemployed.

Q *Thank you Lars. Now can you tell me about your movements on Saturday evening?*

Well, my girlfriend dumped me about a week ago. I blame Hill for a lot of that. He touched me when I was only a kid, and now I just can't keep a steady girl. Anyway, I decided to go to the pub as a distraction, although I really can't afford it.

Q *And you met Steve Nara there? About what time was this?*

Yeah. I arrived about 7, Steve arrived about an hour later. I've met him there a few times now. We have a few mates in common. We had a few drinks together. He's pissed off at Hill too.

Q *So you decided to get even with Hill?*

No, definitely not. I'm not stupid. Anyway, for Hill to pay for what he did to me and my mates, I wouldn't want it over and done so quickly. I wanted him to be publicly humiliated so that everyone, including all his toffy friends, would know what he's really like. That won't happen now.

Q *No, I suppose not. But you did visit the Hill house on Saturday didn't you?*

Yeah, I did. Steve drove me past when we left the pub, and showed me where Hill lived. Then later, on my way home, I stopped the car in front of his house.

Q *What time was this? Where did you park your car?*

About 11 o'clock I suppose. I parked on the nature strip. I walked up his drive feeling pretty bold until the security lights came on, then I hid in some bushes near the front window. I could see Hill asleep on the floor of the front room. I thought he was drunk and had fallen off his chair. I was about to leave when the security lights came on again. There was a guy dressed in dark clothes creeping up the drive. I stayed hidden until he'd walked up to the side door.

Q *So, you watched the man go inside?*

No, I didn't stay to watch. I went back to my car and pushed it onto the street before turning on the engine to drive home. I may have seen his car though. There was a fancy BMW parked in front of a house a few doors up from the Hills' house. It wasn't there when I first drove up.

Q *Can you describe the car? Did you see the man clearly enough to describe him?*

No. It was dark and I was in a hurry to get away. The guy was about average height and a bit older than me I think. I didn't get much of a look at him because I was more interested in getting away without being seen myself. The car was light, white maybe. I suppose that's who you think killed Hill. I'm sorry, I really can't help you with any more. I'm not sorry the guy's gone, but I'd like to have seen him in court first.

Q *Well, that is all for the moment. Thanks for your help Lars. We'll be in touch if we have any further questions to ask you, so don't leave town without letting us know first.*

This statement of 1 pages, signed by me is the truth to the best of my knowledge.

Dated the 22 day of May 20 10 Signed Lars Vogel

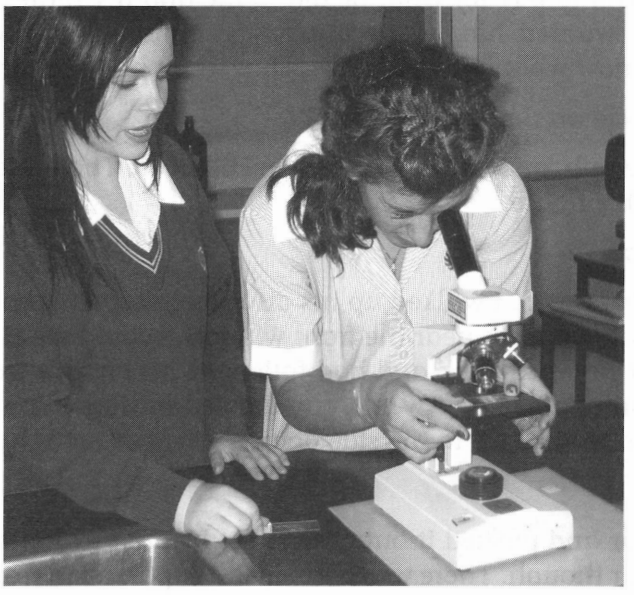
© Blake Education 2006

ISBN 978 1 92561 088 8

Further information from forensic laboratory

- Barnaby Hill's medication had been interfered with. The bottle contained a strong sedative (barbiturate) instead of Barnaby's usual heart tablets. Taking these tablets would have caused Barnaby to lapse into unconsciousness very quickly.
- The cup of tea was exactly that, a cup of tea that had hardly been touched. Only Barnaby's DNA was on the cup.
- The DNA profile of the blood found on Barnaby Hill's car matched the DNA and blood type of Dr Brian McDonald.
- The cuts on the hose were not made with the kitchen knife that Mrs Hill said was hers. She'd used it in the garden to cut twine to tie up her climbing roses. Forensic experts believe the hose was cut with an instrument similar to a scalpel or Stanley knife, also referred to as box cutters. This item has not been located as yet.
- The nurse on duty at the hospital on the night of the murder confirmed that she had spoken to Dr McDonald when he arrived to check on his patient about 10 pm. He told her he had decided to complete some reports to get them out of the way while he was there and had asked not to be interrupted unless there was a serious emergency. He had then gone into his office and the light had been on for a couple of hours. She spoke to him again as he was signing out, just after midnight.
- The hospital log recorded that Dr McDonald arrived at 10.10 pm and left at 12.15 am.

You should now have enough information to arrest at least one of the suspects and charge that person with murder. For the suspect to be committed to trial, there must be enough evidence to convict the person, and this evidence must be strong enough that a lawyer cannot cast doubt in the jury's mind about their guilt.



© Blake Education 2006

ISBN 978 1 92561 088 8

RESTRICTED ACCESS

CONFIDENTIAL

INVESTIGATIVE REPORT: THE DEATH OF BARNABY HILL

INVESTIGATING OFFICER(S): _____

RANK _____ OFFENCE _____

DATE _____ CRIME REFERENCE NUMBER _____

A body is discovered — REPORT PART 1

1 P.C. Paul Anderson told his senior police officer that he believed Barnaby Hill's death was murder, not suicide. Explain what evidence may have led the police officer to this conclusion.

2 Besides Dr Simpson the pathologist (medical examiner), what other support people do you think should be called to the crime scene to help solve the mystery of how Barnaby died?

3 Write a summary of everything you know about the death so far, and what other information you want to find out.

4 Make a list of the people you want to interview. Beside each name, write the questions you would ask them.

The scene is searched at first light — REPORT PART 2

List all the evidence that has been collected so far. Next to each item of evidence, describe what tests or further investigation you think should be carried out on this item.

© Blake Education 2006

ISBN 978 1 92561 088 8

INVESTIGATIVE REPORT

Information obtained by late morning REPORT PART 3

1 List the new evidence that has been discovered. State what should be done with each item of evidence.

2 Summarise all the information you now have about what occurred before Barnaby Hill died.

3 Write a list of suspects. Beside each name, write a motive and reasons why that person is a suspect. Indicate who you think is the most likely murderer, and why.

4 List the information that needs to be looked for, and the tasks that need to be carried out by the investigators.

© Blake Education 2006
ISBN 978 1 92561 088 8

INVESTIGATIVE REPORT

Information obtained by late afternoon REPORT PART 4

Inspect each of the six sets of fingerprints. For each set, record below the pattern on each of the fingers. Use the following codes for the four basic patterns: A = arch, L = loop, W = whorl, C = composite, U = unknown. The more complex patterns have the following codes: UL = ulna loop (slopes down towards the little finger), RL = radial loop (slopes down towards the thumb), TA = tented arch, DL = double loop and CPL = central pocket loop. Write S to indicate scars.

Suspect or victim	Left hand					Right hand				
	little finger	ring finger	middle finger	fore-finger	thumb	thumb	fore-finger	middle finger	ring finger	little finger
Barnaby Hill										
	Any other comments regarding palms, fingers or size of hands									
Helena Hill										
	Any other comments regarding palms, fingers or size of hands									
Dr Brian McDonald										
	Any other comments regarding palms, fingers or size of hands									
Isabella McDonald										
	Any other comments regarding palms, fingers or size of hands									
Timothy Hill										
	Any other comments regarding palms, fingers or size of hands									
Sarah Hill										
	Any other comments regarding palms, fingers or size of hands									

Crime scene fingerprints: Analyse the prints found. Use the codes as described above to record the patterns. Use X to indicate that a fingerprint is not present. Use ? to indicate that the pattern is not clear. Compare these with the fingerprint analysis that you recorded in the table above.

Prints	Left hand					Right hand				
	little finger	ring finger	middle finger	fore-finger	thumb	thumb	fore-finger	middle finger	ring finger	little finger
On glass cabinet										
	Any other comments regarding palms, fingers or size of hands									
On front car door										
	Any other comments regarding palms, fingers or size of hands									
On back car door										
	Any other comments regarding palms, fingers or size of hands									
On knife										
	Any other comments regarding palms, fingers or size of hands									

1 Record
- time and possible cause of death
- whose fingerprints were found where and if this is significant or not
- which person's handwriting matches that on the two notes and why you think they match.

© Blake Education 2006

ISBN 978 1 92561 088 8

2 Revise your list of suspects, giving reasons why you think each might be the murderer.

Some practical activities REPORT PART 5

1 Record whose pens matched the ink used in the two notes.

Crime scene suicide note: _____ Warning note: _____

2 Does this prove beyond all reasonable doubt that the owners of the pens wrote the notes? Explain who you think wrote each note, and why.

3 What other information would be useful in confirming who wrote the notes?

4 Record which, if any, of the fibres match those from the crime scene. Discuss how significant this might be for the case. That is, is there a simple explanation for why the owner of the garment left these fibres? Or does it provide strong evidence that may be used to convict the murderer?

5 Record which, if any, of the hair samples match those found in the seat belt of Barnaby Hill's car. Discuss the significance or strength of this evidence.

© Blake Education 2006
ISBN 978 1 92561 088 8

INVESTIGATIVE REPORT

6 Revise your list of suspects. In each case, say why you think that this person might be the murderer.

7 Make a list of any other people you wish to interview and the questions you would like to ask them.

Information available early next day ·· REPORT PART 6

1 Summarise any new information provided by the statements.

2 Record whose shoe prints and tyre prints were found at the crime scene. Discuss the significance of this to the solving of the crime.

3 Revise your list of suspects in light of this new information.

4 Who would you still like to interview? What questions would you ask them?

© Blake Education 2006

ISBN 978 1 92561 088 8

INVESTIGATIVE REPORT

Final evidence from a suspect and the lab .. REPORT PART 7

The death of Barnaby Hill

Record the name of the person you are charging with the murder of Barnaby Hill, and their motive. Write a summary of how the crime was committed. Describe when it was planned, how it was carried out (including the times) and the evidence you have that shows that this person is the murderer beyond all reasonable doubt.

© Blake Education 2006
ISBN 978 1 92561 088 8

Answers

Activity 1: Recording shoe prints and tyre marks

1. No, the tread pattern of the impression is quite different to that on the shoe.
2. They are permanent records that can be referred to once the original marks have been removed. This is especially important during a court case when a jury can see an exact 3D copy of the original object.
3. Both are made in case one is less clear.
4. A matching shoe print does not prove that the owner committed the crime or was wearing the shoe at the time. Courts require proof 'beyond all reasonable doubt'.
5. The more evidence that incriminates a suspect, the stronger the case against them. Information such as where the suspect was at the time, as well as fingerprints or DNA that individualises evidence.

Activity 2: Inspecting hairs and fibres

1. The type of animal and even which part of the body it came from, if it was pulled out, cut or dropped out naturally, hair treatments used and what drugs or food the person ingested. It may contain enough DNA to profile.
2. It may mean that the criminal owns or has contact with rabbits or wears clothes made from rabbit fur.
3. a. Natural and synthetic. b. Natural fibres (animal) – angora (goat), alpaca (wool from alpaca), mohair (goat), (plant) – linen (flax). Synthetic fibres – polyamide, vinyl, acrylic.
4. Natural fibres are inspected under a microscope and may undergo chemical analysis for poisons or drugs. Synthetic fibres will be inspected under a microscope, usually with light of different wavelengths. They maybe tested for strength, melting point or chemical analysis to determine their chemical makeup.
5. No, the investigator would need to know who was wearing the clothes at the time, how common the fibres were, and where the suspect was at the time the crime was committed.

Activity 3: Using chromatography to compare inks

1. A chromatogram is made when some ink or dye is placed near the bottom of some absorbent paper and the paper is stood in a solvent, which will carry the ink with it as the solvent soaks up the paper.
2. Ink is made up of different chemicals, usually of different colours. The smaller or more soluble particles move faster, so are carried further upwards than the larger or less soluble ones.
3. If two chromatograms are identical, it suggests that the same chemical mix was in each ink. The actual pens may be different but are probably of the same type and produced by the same manufacturer.
4. Yes. The colours will spread out more, but always in the same order, if left in the solvent longer.
5. No. Even if it is shown the same pen was used, it does not determine who used the pen. Handwriting comparison would be useful evidence if trying to determine who wrote the note.
6. No. Today scientists use instruments that identify each chemical and the amount of each in the mixture.

Part 1: A body is discovered

1. Barnaby Hill had bare feet which were clean, indicating he probably did not walk outside, but may have been carried or dragged out.
2. Crime Scene Examiners to search the house and yard, police for security and to question neighbours.
3. Barnaby Hill is dead, possibly from gas poisoning; not accidental (clean feet, lost slippers), garden hose placed on car to cause death. No clear suspects at this stage. You would ask the family and friends if he had a reason to commit suicide or had any enemies who might want to kill him.
4. Patrick Sullivan – how well he knew Barnaby and his family, exactly what he knew about Barnaby's last hours and if he'd noticed anything strange recently. Barnaby's family, friends & associates – their movements, reasons why Barnaby might kill himself or be killed.

Part 2: The scene is searched at first light

- 3 m of garden hose, plus Hill's garden hose – collect, see if pieces match and if knife was used.
- Cloth tied to the 3 m garden hose – collect, try to find source.
- Dark blue fibres – collect and take to lab for analysis and comparison to fibres collected from suspects.
- Knife – collect and send to lab to develop fingerprints and compare to cuts on hose.
- Dark brown stain, presumed blood – collect, send to lab for blood typing and DNA profiling.
- Two long brown hairs – collect, compare to hair from suspects.
- Shoe prints and tyre prints – photograph with side light, make a cast, compare to suspects' tyres and shoes.
- All fingerprints – dust, photograph, lift with sticky tape, send to lab for matching.

Part 3: Information obtained by late morning

1. • Heart tablets - collect, send bottle to lab for fingerprinting and chemical analysis of tablets.
   • Cup of tea - collect liquid and send to lab for analysis, test cup for fingerprints and DNA from saliva.
   • Second note – collect, send to lab for handwriting comparison, fingerprints, chromatography for ink.
2. 7–9.30 pm – the family (Barnaby, his wife Helena, daughter Sarah, son Tim and his fiancée, Amy) had dinner with friends Dr and Mrs McDonald. 8 pm – fencer Steve Nara arrives to ask for payment. 9.30 pm - Dr and Mrs McDonald left. 9.45 pm – Sarah left, Helena went to bed. 10 pm – Tim and Amy left. 3.30 am – Patrick discovered body. Barnaby died between 10 pm and 3 am.
3. Suspects: Steve Nara, the fencer, is owed money by Barnaby; Lars Vogel, who has accused Barnaby of molesting him. Family members who might inherit from Barnaby's will and were the last to see him.
4. All family, friends and suspects, should be interviewed and their movements (alibis) recorded. They need to give samples of their fingerprints, handwriting, blood and DNA, clothes and shoes they were wearing, and tyre patterns of their cars.

© Blake Education 2006

ISBN 978 1 92561 088 8

Part 4: Information obtained by late afternoon

Barnaby Hill	UL	UL(S)	A	A	DL	DL	CPL	UL	UL	UL
	Scar on left ring finger									
Helena Hill	UL	UL	A	UL	UL	UL	A	A	UL	UL
	Lots of fine cracks or fissures on all fingers									
Dr Brian McDonald	UL	UL	UL	RL	DL	W	RL	UL	UL	UL
Isabella McDonald	UL	UL	W	W	UL	UL	W	UL	UL	UL
Tim Hill	CPL	W	W	RL	CPL	W	W	UL	W	UL
Sarah Hill	UL	UL	TA	A	UL	A	A	UL	UL	UL
On glass cabinet	UL	UL(S)	A	A	?	X	X	X	X	X
	Scar on left ring finger, so prints match Barnaby Hill									
On front car door	X	X	X	X	X	A	A	UL	?	X
	Fingerprint matches Sarah Hill's right hand									
On back car door	X	UL	UL	RL	DL	X	X	X	X	X
	Not sure which hand thumbprint is from, probably left. Prints match Dr McDonald's left hand									
On knife	X	X	UL	UL	A	X	X	X	X	X
	Prints difficult to match, but the cracks and fissures match those of Helena Hill									

1 • **Time of death:** 11.30 pm–midnight. **Cause:** barbiturates – unconsciousness, carbon monoxide killed him.
• **Fingerprints:** glass cabinet = Barnaby Hill, front car door = Sarah Hill, back of car = Dr Brian McDonald, knife = Helena Hill. Only the doctor's are suspicious, Sarah borrowed the car.
• **Suicide note** closest to Barnaby Hill's handwriting, **second note** matches Isabella McDonald's writing. Both notes match in size of letters, spacing, slope and shape of certain letters.

2 Dr McDonald is now a suspect, his fingerprints are on the car near the exhaust pipe. No obvious explanation at this stage and he has access to barbiturates. Second note suggests Isabella is having an affair with Barnaby.

Part 5: Some practical activities

1 Suicide note matches Helena's pen. The other note matches Isabella's pen, strengthening the suggestion that she was having an affair and warned Barnaby that Brian had found out and might harm him.

2 Anyone with access to the house could have used Helena's pen to write the suicide note. The murderer may have forged Barnaby's handwriting using Helena's pen to make it look like suicide. Isabella had no need to disguise her note.

3 Handwriting comparisons, fingerprints or DNA left on notes, context of message and who was recipient.

4 Blue fibres match those from Dr McDonald's jumper – no logical explanation. He is main suspect now.

5 Long brown hair strands match those from Sarah Hill – she borrowed Barnaby's car for several days.

6 Dr McDonald – his fingerprints are on the car and his fibres on the cloth. Motive: Barnaby's affair with his wife. Other suspects as before.

7 Mrs McDonald, Steve Nara, Lars Vogel – ask about their relationships with Barnaby and their movements at the time of the murder.

Part 6: Information available early next day

1 Helena had an affair with Barnaby that Brian was angry about (motive). No one can confirm that she was not at the crime scene at the time of the murder. Warning Barnaby to be careful in her note makes it unlikely that she murdered him. Steve Nara had an alibi for the early part of the evening, but not after 11 pm.

2 Steve Nara's shoe print near side door – probably made during his 8 pm visit, Lars Vogel's shoe print at front of house – no explanation, Tim Hill's tyre prints on front lawn where he parked during the dinner party, Lars Vogel's tyre print, the closest to those found on the nature strip, is suspicious.

3 Dr McDonald is still the main suspect, although Lars Vogel's tyre and shoe prints need explanation.

4 Lars Vogel regarding his movements and explanation for shoe and tyre prints.

Part 7: Final evidence from a suspect and the lab

The murderer is Dr Brian McDonald: motive, to stop the affair between his wife and Barnaby. He was angry when he realised the affair continued after Isabella had promised to stop. He collected barbiturates from his car and swapped them with Barnaby's heart tablets when he helped Helena get the coffee, using a handkerchief to prevent leaving fingerprints. As a doctor, he knew Barnaby's tablet routine, and that the drugs would render him unconscious before 11 pm. Lars Vogel's description of man and car fit the doctor, but are not conclusive. Dr McDonald went to the hospital to create an alibi, making sure he was seen when he arrived. He turned his office light on and signed the log, leaving at 12.15 am. However, he really left just after 10.30, giving him time to get to Barnaby's house about 11 pm. He found Barnaby unconscious and fallen off his chair. Using a pen he found in the kitchen, the doctor wrote the suicide note, forging Barnaby's handwriting. He then dragged Barnaby to his car, the slippers coming off as he went. He set up the exhaust using an old cloth that he had in the boot of his car, and cut the garden hose with an old scalpel that he later threw away. He cut his hand and removed his gloves to put on a bandaid, leaving fingerprints on the car. He returned to the hospital, making sure he was seen again by the same nurse, signing out at just after midnight.

© Blake Education 2006
ISBN 978 1 92561 088 8

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing **the highest quality learning outcomes with instant usability.**

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Forensic Science Hypothetical: The Death of Barnaby Hill

This second title in the **Forensic Science Hypothetical** series by best-selling author Jeanette Jolley provides photocopiable worksheets containing all the information and evidence that students need to solve a crime involving a death. Students apply their science skills to a real world example of crime, and teachers can thoroughly assess what their class has learnt through an extended hands-on task. Students will enjoy investigating who committed the crime using key elements of forensic science such as collecting and examining evidence; analysing hairs, fibres and handwriting; and eliminating suspects.

No previous study or knowledge of forensic science is necessary as the first three activities of the book are designed to provide background information for students who have no formal knowledge of forensic science, or need to revise these skills. The activities require minimal teacher preparation and answers have been provided.

Topics include

- Crime scenes and the examination of them
- Interpreting maps
- Analysing witness and suspect statements
- Exploring motives
- Recording and comparing shoe prints and tyre marks
- Inspecting hairs, fibres and handwriting evidence
- The skills of taking fingerprints and eliminating suspects
- Interpreting a post mortem report
- Testing inks with chromatography
- Forensic laboratories

Associated titles in the Instant Lessons series

- Forensic Science Hypothetical
  - The Kidnapping of Jamie Lawson
  - The Burglary in Sutton Street
- Forensic Science for High Schools (Books 1, 2, 3, 4)
- Literacy in Science (Books 1, 2, 3, 4)
- Legends of Science (Books 1, 2)

Blake EDUCATION
Better ways to learn

Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

ISBN: 978-1-92561-088-8
9 781925 610888